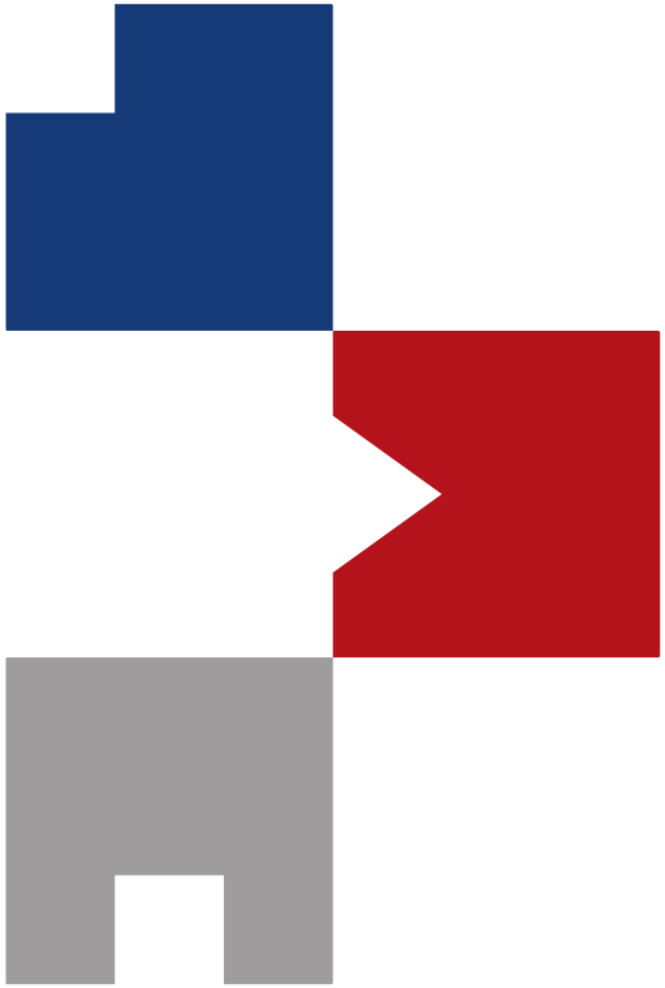
Note) Slides and example codes are available:
https://github.com/mint-lab/3dv_tutorial



An Invitation to 3D Vision: Finding Correspondences

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Correspondence Problem

- Q) What are the three most important problems in computer vision?
- Takeo Kanade) Correspondence, correspondence, and correspondence!

September 2019

Learning and Reasoning with Visual Correspondence in Time

View Publication

Authors:
Xiaolong Wang

PhD Thesis | Technical Report | CMU-RI-TR-19-75

Abstract:

There is a famous tale in computer vision: Once, a graduate student asked the famous computer vision scientist Takeo Kanade: "What are the three most important problems in computer vision?" Takeo replied: "Correspondence, correspondence, correspondence!" Indeed, even for the most commonly applied Convolutional Neural Networks (ConvNets), they are internally learning representations that lead to correspondence across objects or object parts. The way these networks learn is via human annotations on millions of static images. For example, humans will label images as dog, car, etc. However, this is not how we humans learn. The visual system of an infant develops in a dynamic and continuous environment without using semantics until much later in life.

In this thesis, I will argue that we need to go beyond images and exploit the massive amount of correspondence in videos. In videos, we have millions of pixels linked to each other by time. I will discuss how to learn correspondence from continuous observations in videos without any human supervision. Once the correspondence is given, it can be utilized as supervision in training the ConvNets, eliminating the need for manual labels. Besides supervision, capturing long-range correspondence is also the key to video understanding as well as interaction reasoning. The effectiveness of these ideas will be demonstrated on tasks including object recognition, tracking, action recognition, affordance and physical property estimation.

Notes:

Show BibTeX Copy BibTeX



Takeo Kanade

Image: CMU Robotics Institute, Kyoto University

Table of Contents: Finding Point Correspondences

- **Feature Points**
  - Q) How can we detect salient points (or parts)?
- **Feature Descriptors**
  - Q) How can we distinguish the points from each other?
- **Feature Matching**
  - Q) How can we associate the points across different images?
- **Feature Tracking**
  - Q) How can we associate the points across their next image?
- **Outlier Rejection**
  - Q) How can we select correctly matched points?



Image: pixabay

Getting Started with a Quiz

- **Q) What is it?**



600 pixels

Getting Started with a Quiz

- **Q) What is it?**



600 pixels

Getting Started with a Quiz

- **Q) What is it? "Duck"**



600 pixels 1095 pixels 600 pixels

- **Q) Why corners (junction) instead of edges or blobs?**
  - The human visual system understands objects better from corners than from edges.
  - a.k.a. *feature* points, *key*points, *salient* points, and *interest* points

Getting Started with a Quiz

- **Q) Why corners (junction) instead of edges or blobs? Where are the image patches?**

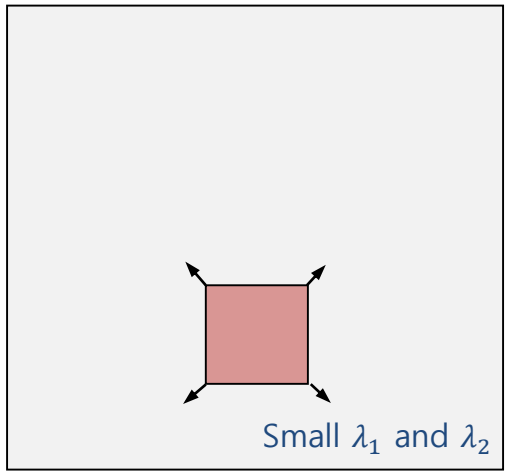


- **Q) What is a good feature? (Requirements)**
  - **Repeatability** (to be invariant/robust to transformations and noise)
  - **Distinctiveness** (to be easy to distinguish or match)
  - **Locality** (due to occlusion)
  - Quantity (sufficient number), accuracy (localization), efficiency (computing time), …

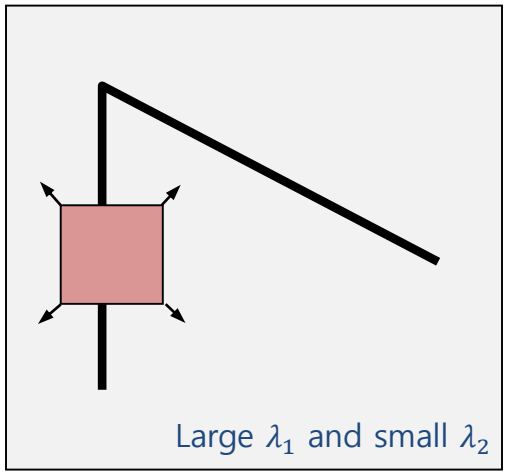
Image: OpenCV Tutorials

Feature Point) Harris Corner (1988)

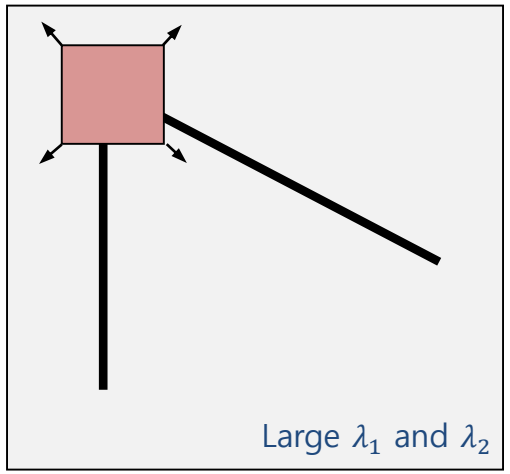
- Key idea: **Sliding window**



"flat" region:
no change
in all directions



"edge":
no change
along the edge direction



"corner":
significant change
in all directions

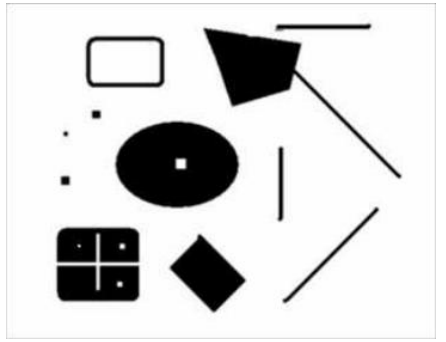
– Formulation

- $I(x+\Delta_x, y+\Delta_y) \approx I(x,y) + [I_x(x,y) \quad I_y(x,y)] \begin{bmatrix}\Delta_x \\ \Delta_y\end{bmatrix}$ where $I_x = \frac{\partial I}{\partial x}$ and $I_y = \frac{\partial I}{\partial y}$
- $D(\Delta_x, \Delta_y) = \sum_{(x,y)\in W} \left(I(x+\Delta_x, y+\Delta_y) - I(x,y)\right)^2 \approx [\Delta_x \quad \Delta_y] \underbrace{\begin{bmatrix}\sum_W I_x^2 & \sum_W I_x I_y \\ \sum_W I_x I_y & \sum_W I_y^2\end{bmatrix}}_{\mathrm{M}} \begin{bmatrix}\Delta_x \\ \Delta_y\end{bmatrix}$
- Two eigenvalues of M: λ_1 and λ_2

Image: Frolova and Simakov, "Matching with Invariant Features" (Lecture Slides), 2004

Feature Point) Harris Corner (1988)

$I(x, y)$

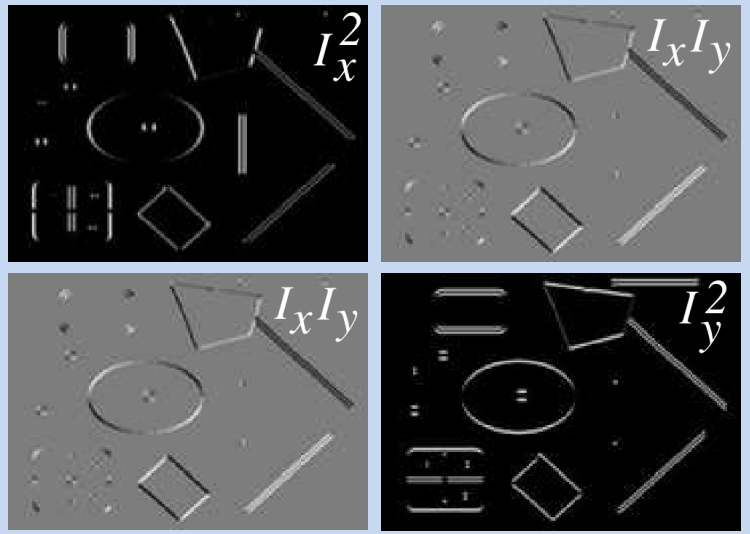


- Key idea: **Sliding window**
  - Formulation
    - $I(x + \Delta_x, y + \Delta_y) \approx I(x, y) + [I_x(x, y) \quad I_y(x, y)] \begin{bmatrix} \Delta_x \\ \Delta_y \end{bmatrix}$ where $I_x = \frac{\partial I}{\partial x}$ and $I_y = \frac{\partial I}{\partial y}$
    - $D(\Delta_x, \Delta_y) = \sum_{(x,y) \in W} \left(I(x + \Delta_x, y + \Delta_y) - I(x, y) \right)^2 \approx [\Delta_x \quad \Delta_y] \begin{bmatrix} \sum_W I_x^2 & \sum_W I_x I_y \\ \sum_W I_x I_y & \sum_W I_y^2 \end{bmatrix} \begin{bmatrix} \Delta_x \\ \Delta_y \end{bmatrix}$

M

  - **Harris corner response**
    - $\text{cornerness} = \det(\text{M}) - k \, \text{trace}(\text{M})^2$
    - Note) $\det(\text{M}) = \lambda_1 \lambda_2$, $\text{trace}(\text{M}) = \lambda_1 + \lambda_2$, and $k \in [0.04, 0.06]$
  - Note) **Good-Feature-to-Track** (a.k.a. GFTT or Shi-Tomasi corner; 1994)
    - $\text{cornerness} = \min(\lambda_1, \lambda_2)$

M



$\det(\text{M}) - k \, \text{trace}(\text{M})^2$



Image: Tuytelaars et al., "Local Invariant Feature Detectors: A Survey", Foundations and Trends in Computer Graphics and Vision, 2008 DOI

Feature Point) Harris Corner (1988)

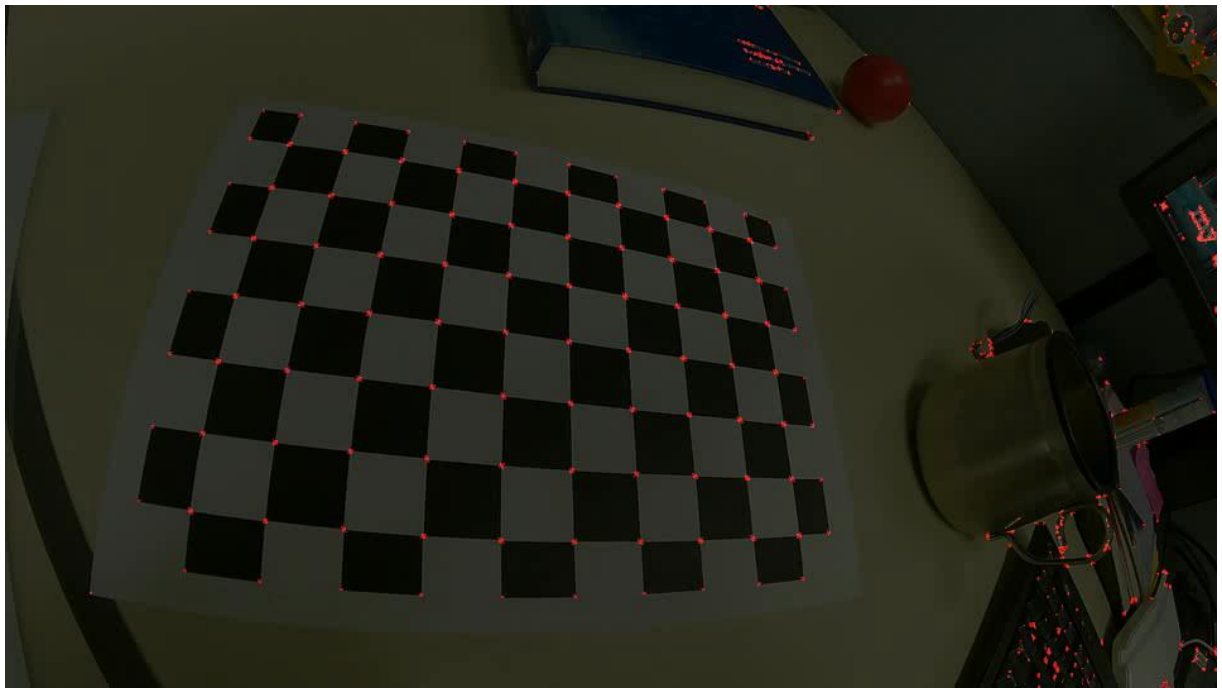
- Example) **Harris corner implementation** [harris_corner_implement.py]

```
def cornerHarris(img, ksize=3, k=0.04):
    # Compute gradients and M matrix
    Ix = cv.Sobel(img, cv.CV_32F, 1, 0)
    Iy = cv.Sobel(img, cv.CV_32F, 0, 1)
    M11 = cv.GaussianBlur(Ix*Ix, (ksize, ksize), 0)
    M22 = cv.GaussianBlur(Iy*Iy, (ksize, ksize), 0)
    M12 = cv.GaussianBlur(Ix*Iy, (ksize, ksize), 0)

    # Compute Harris cornerness
    detM = M11 * M22 - M12 * M12
    traceM = M11 + M22
    cornerness = detM - k * traceM**2
    return cornerness
```

$$\mathrm{M} = \begin{bmatrix} \sum_W I_x^2 & \sum_W I_x I_y \\ \sum_W I_x I_y & \sum_W I_y^2 \end{bmatrix}$$

$$\text{cornerness} = \det(\mathrm{M}) - k\,\mathrm{trace}(\mathrm{M})^2$$



Feature Point) Harris Corner (1988)

- Properties
  - Invariant to translation, rotation, and intensity shift ($I \rightarrow I + b$) ~~intensity scaling ($I \rightarrow aI$)~~
  - Variant to **image scaling**



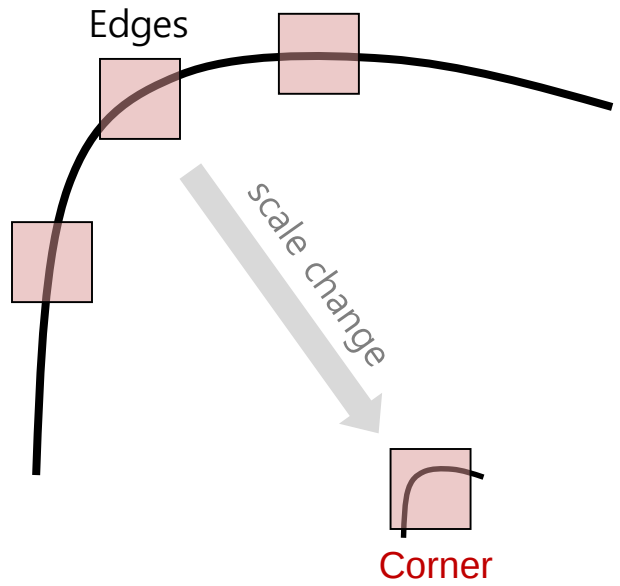
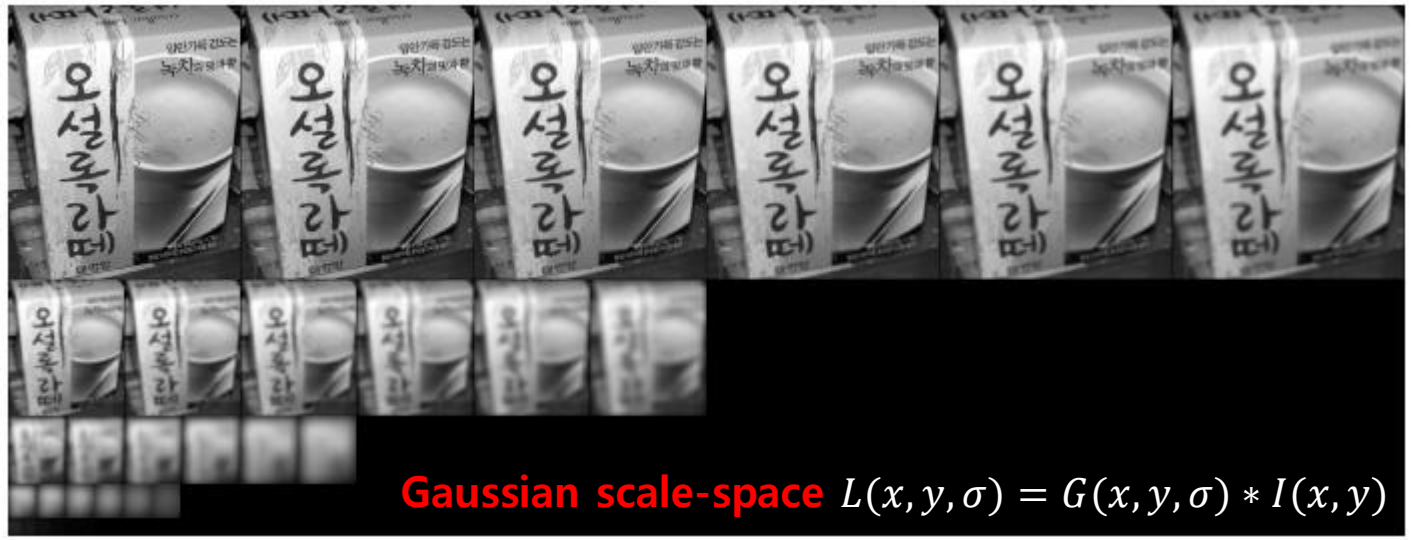


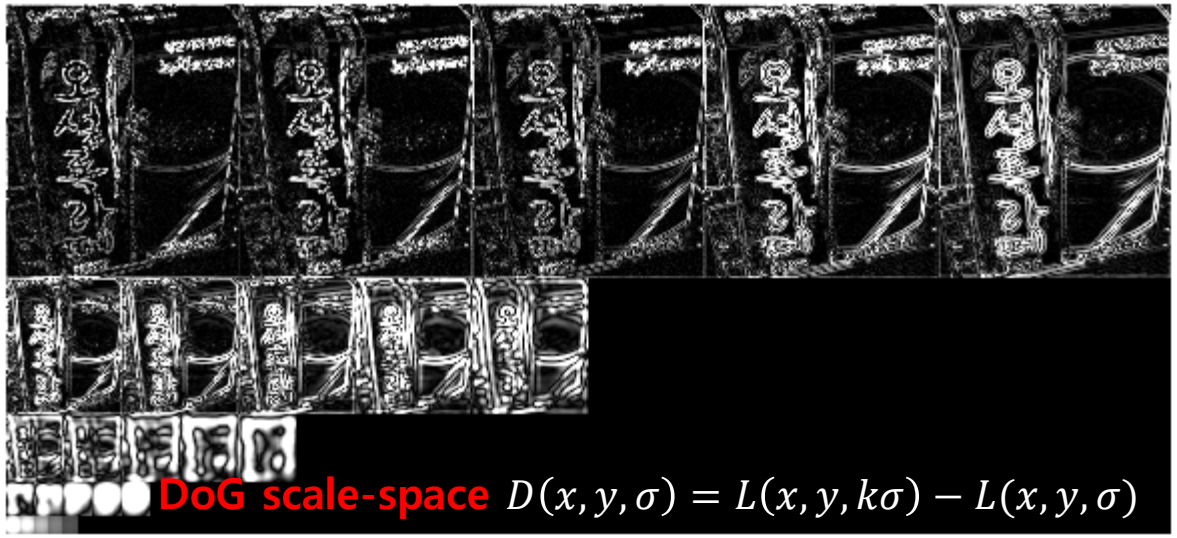
Image: Frolova and Simakov, "Matching with Invariant Features" (Lecture Slides), 2004

Feature Point) SIFT (Scale-Invariant Feature Transform; 1999)

- Key idea: **Scale-space** (~ image pyramid) **and DoG** (difference of Gaussian)





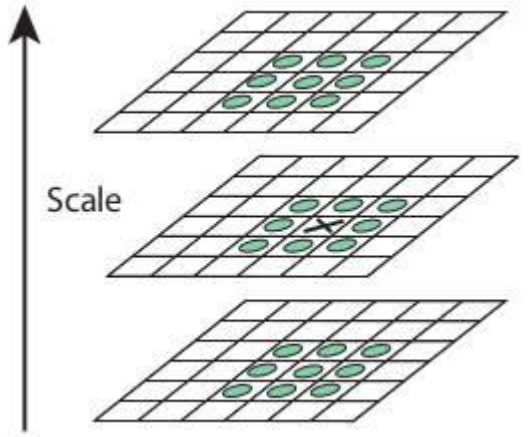


Note) The patent for SIFT feature and descriptors has expired since March 6, 2020.

Feature Point) SIFT (Scale-Invariant Feature Transform; 1999)

- Key idea: **Scale-space** (~ image pyramid) **and DoG** (difference of Gaussian)
- Part #1) **Feature point detection**
  1. Find **local extrema** (minima and maxima) in DoG scale-space
  2. Localize their position accurately (sub-pixel level) using 3D quadratic function
  3. Eliminate **low contrast candidates**, $|D(\mathbf{x})| < \tau$
  4. Eliminate **candidates on edges**,

$$\frac{\text{trace}(H)^2}{\det(H)} < \frac{(r+1)^2}{r} \quad \text{where} \quad H = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix}$$



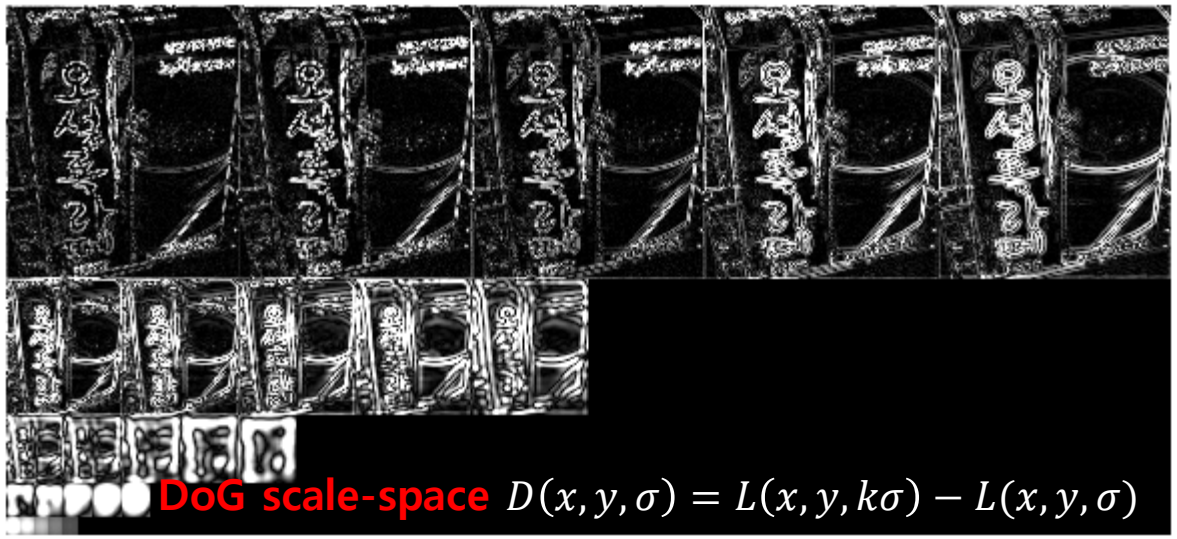
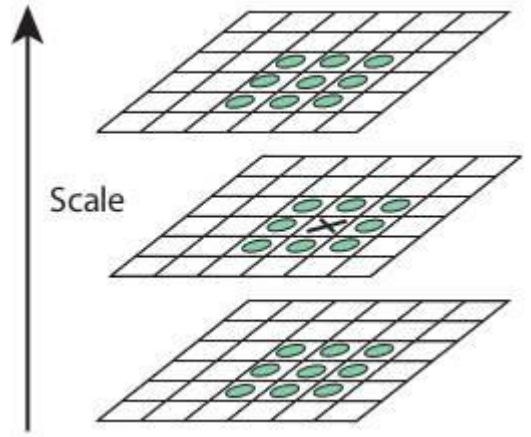


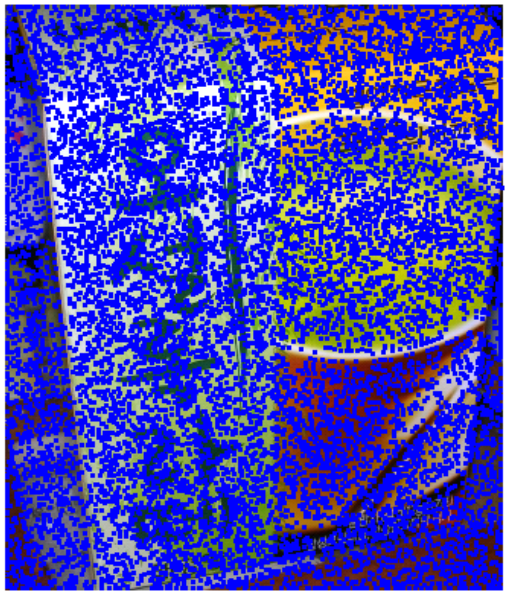
Image: David Lowe, "Distinctive Image Features from Scale-invariant Keypoints", IJCV, 2004 DOI PDF

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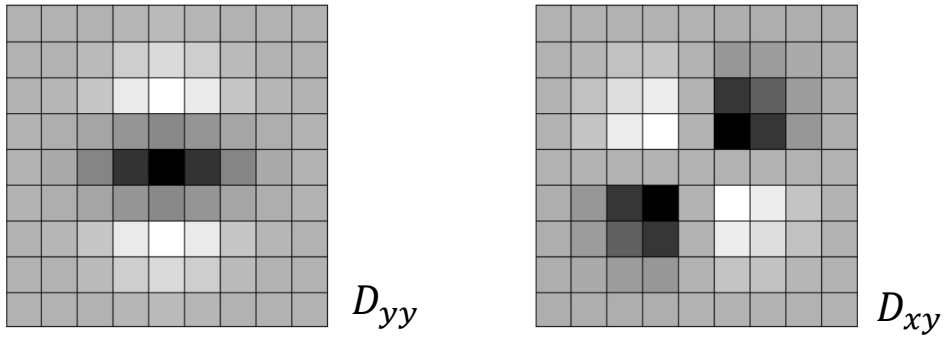
Local extrema (N: 11479)

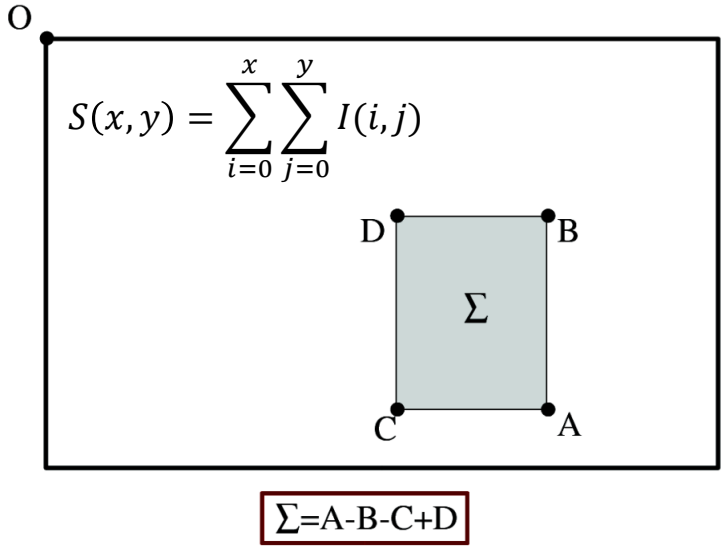


Feature points (N: 971)

Feature Point) SURF (Speeded Up Robust Features; 2006)

- Key idea: **Approximation of SIFT**
  - e.g. DoG approximation Haar-like features and **integral image**





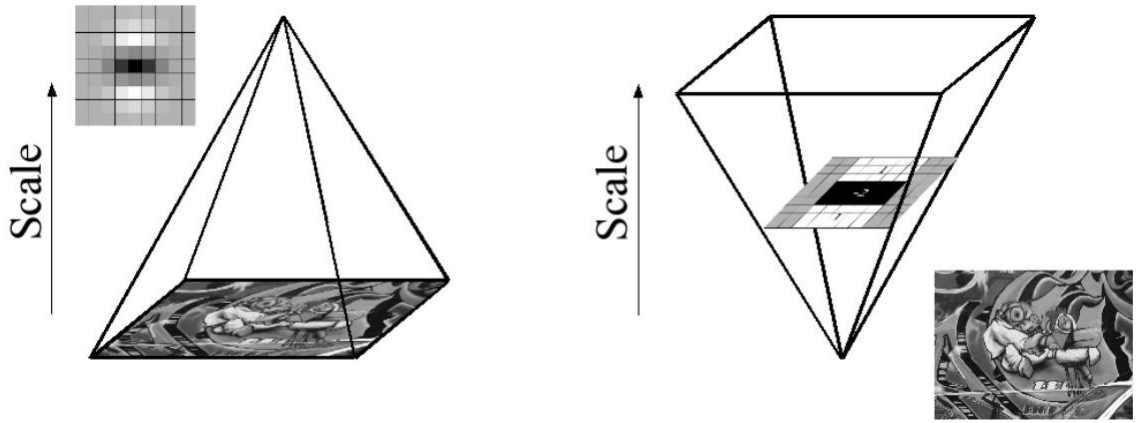


Image: Bay et al., "Speeded Up Robust Features (SURF)", CVIU, 2008 DOI

Feature Descriptor) SIFT (Scale-Invariant Feature Transform; 1999)



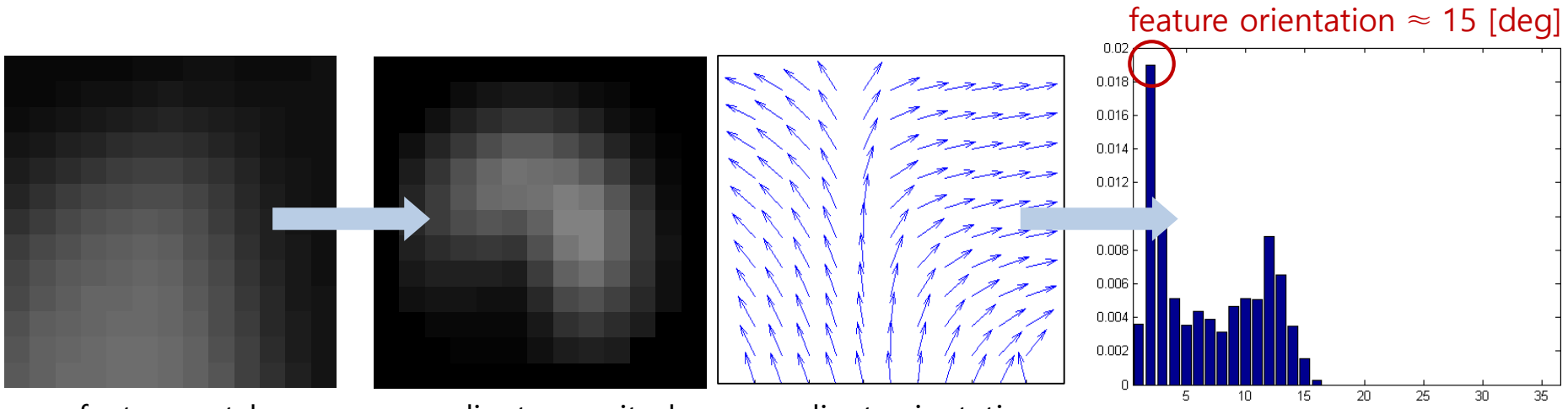
Feature scales and orientations

- Part #2) **Orientation assignment**
  1. Derive magnitude and orientation of gradient of each patch

$$m(x,y) = \sqrt{\big(L(x+1,y) - L(x-1,y)\big)^2 + (L(x,y+1) - L(x,y-1))^2}$$

$$\theta(x,y) = \tan^{-1}\frac{L(x,y+1) - L(x,y-1)}{L(x+1,y) - L(x-1,y)}$$

  2. Find the strongest orientation
     - Histogram voting (36 bins) with Gaussian-weighted magnitude



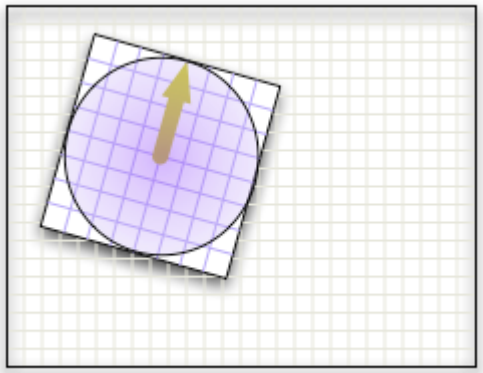
feature patch | gradient magnitude $m(x,y)$ | gradient orientation $\theta(x,y)$ | histogram voting (36 bins)

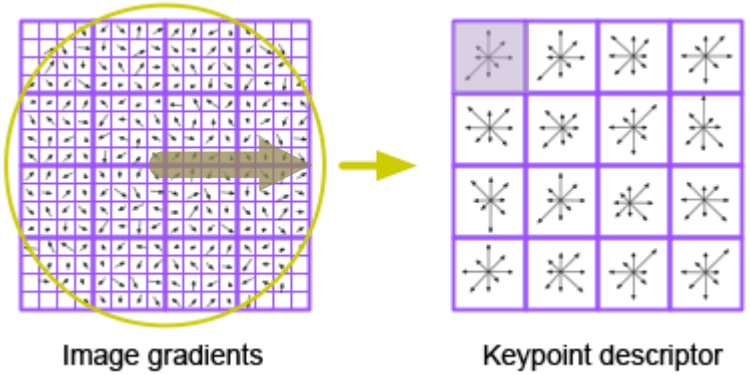
Feature Descriptor) SIFT (Scale-Invariant Feature Transform; 1999)

- Part #3) **Feature descriptor extraction**
  1. Build a 4x4 gradient histogram (8 bins) from each patch (16x16 pixels)
     - Use Gaussian-weighted magnitude again
     - Use relative angles w.r.t. the assigned feature orientation
  2. Encode the histogram into a 128-dimensional vector
     - Normalize it to be an unit vector



Feature scales and orientations





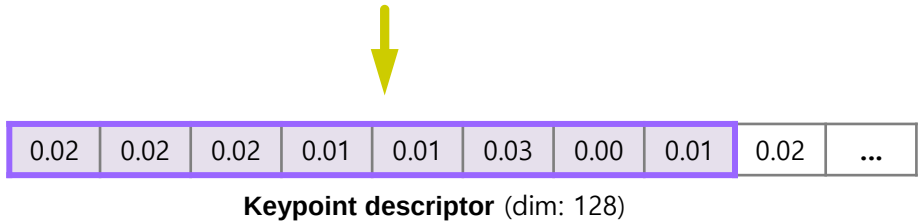
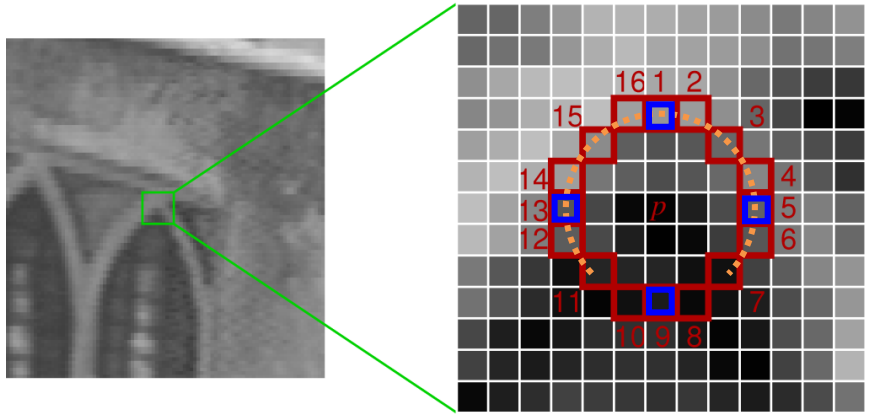


Image: Jonas Hurrelmann, "Harris Corner Detector & SIFT" (Lecture Slides), 2006

Feature Point) FAST (Features from Accelerated Segment Test; 2006)

- Key idea: **Intensity check of continuous arc of n pixels**
  - Is this point p a corner? (I_p: intensity at p, t: the intensity threshold)
    - Is a segment of n continuous pixels brighter than $I_p + t$? (OR) Is the segment darker than $I_p - t$?
    - Note) High-speed non-corner rejection: Checking intensity values at 1, 9, 5, and 13 (n: 12)



  - Too many corners! → Non-maximum suppression
- Versions
  - FAST-9 (n: 9; `cv.FastFeatureDetector_TYPE_9_16`), FAST-12 (n: 12), …
  - FAST-**ER**: Training a decision tree to **e**nhance **r**epeatability with more pixels

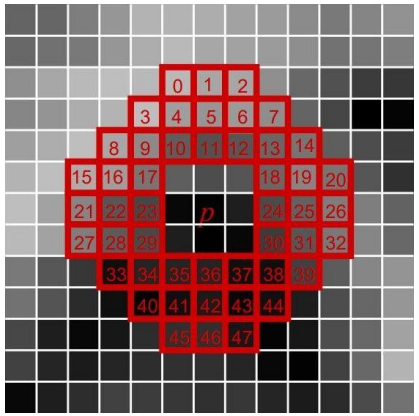
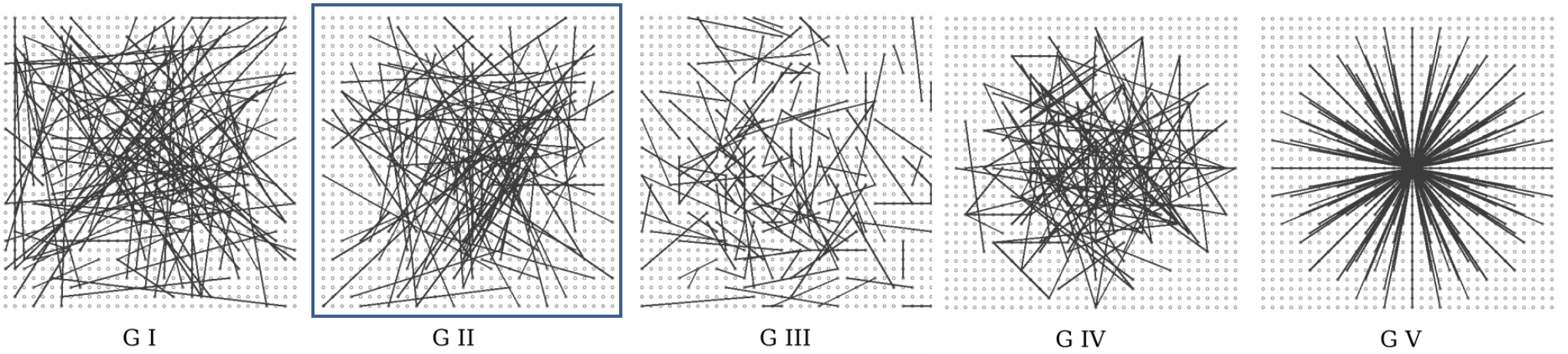


Image: Rosten et al., "Faster and Better: A Machine Learning Approach to Corner Detection", T-PAMI, 2010 DOI

Feature Descriptor) BRIEF (Binary Robust Independent Elementary Features; 2010)

- Key idea: **Intensity comparison of a sequence of random pairs (binary test)**
  - Patch size: 31 x 31 pixels (Note: Applying smoothing for stability and repeatability)



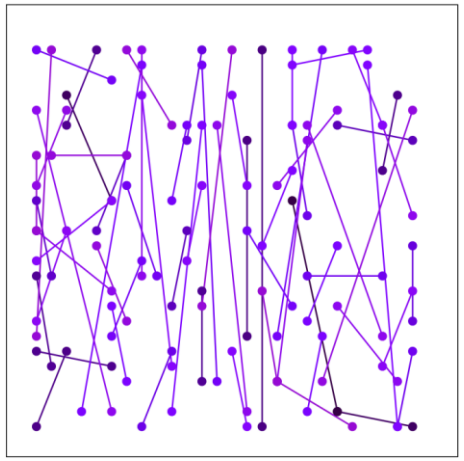
  - Descriptor size: 128 tests (128 bits) → 16 bytes
    - Note) SIFT: 128-dimensional vector → 512 bytes
- Versions: The number of tests
  - BRIEF-32, BRIEF-64, BRIEF-128 (16 bytes), BRIEF-256 (32 bytes), BRIEF-512 (64 bytes), …
- Combination examples
  - SIFT feature points + BRIEF descriptors
  - FAST feature points + BRIEF descriptors

Image: Calonder et al., "BRIEF: Computing a Local Binary Descriptor Very Fast", T-PAMI, 2012 DOI PDF

Feature Point and Descriptor) ORB (Oriented FAST and rotated BRIEF, 2011)

- Key idea: **Adding rotation invariance to BRIEF**
  - **Oriented FAST**
    - Generate scale pyramid for scale invariance
    - Detect *FAST-9* points (filtering with Harris corner response)
    - Calculate feature orientation by *intensity centroid* $C = (\frac{m_{10}}{m_{00}}, \frac{m_{01}}{m_{00}})$

      $\theta = \tan^{-1}\frac{m_{01}}{m_{10}}$ where $m_{pq} = \sum_{x,y} x^p y^q I(x,y)$
  - **Rotation-aware BRIEF**
    - Extract BRIEF descriptors w.r.t. the known orientation
    - Use better comparison pairs trained by greedy search



- Combination (default): **ORB**
  - FAST-9 detector (with orientation) + BRIEF-256 descriptor (with trained pairs)
- Computing time
  - ORB: **15.3 [msec]** / SURF: 217.3 [msec] / SIFT: 5228.7 [msec] @ 24 images (640x480) in Pascal dataset

Image: Rublee et al., "ORB: An Efficient Alternative to SIFT or SURF", ICCV, 2011 DOI

Feature Point and Descriptor) LIFT (Learned Invariant Feature Transform; 2016)

- Key idea: **Deep neural network**
  - CNN network: **DET** (feature detector) + **ORI** (orientation estimator) + **DESC** (feature descriptor)
  - Training data: Photo Tourism dataset with VisualSFM (SIFT)

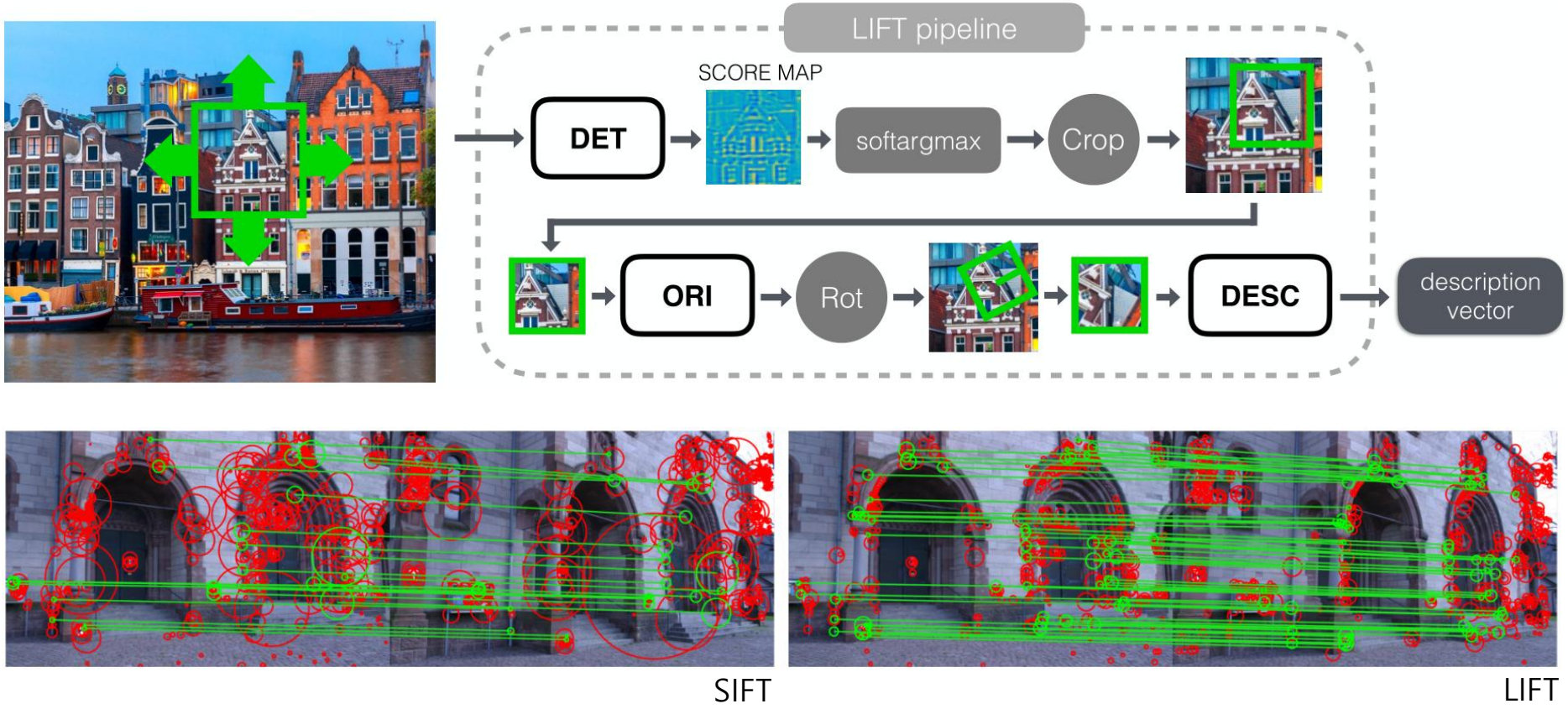


Image: Yi et al., "LIFT: Learned Invariant Feature Transform", ECCV, 2016 arXiv Github

Feature Point and Descriptor) SuperPoint (2017)

- Key idea: **Self-supervised training with homography transformation**
  - CNN network: Encoder (~ VGG) + Decoders (for point and descriptor)

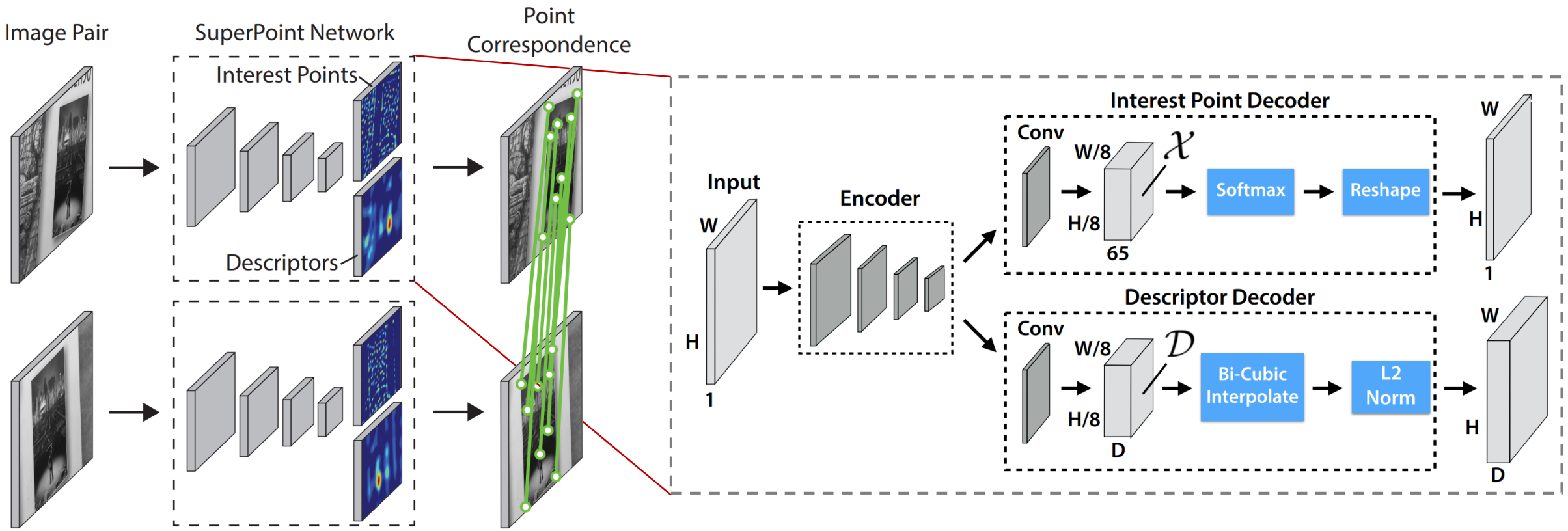


Image: DeTone et al., "SuperPoint: Self-Supervised Interest Point Detection and Description", CVPRW, 2018 arXiv Github (authors) Github

Feature Point and Descriptor) SuperPoint (2017)

- Key idea: **Self-supervised training with homography transformation**
  - Training data generation: The base detector, *MagicPoint* → Ground truth interest points

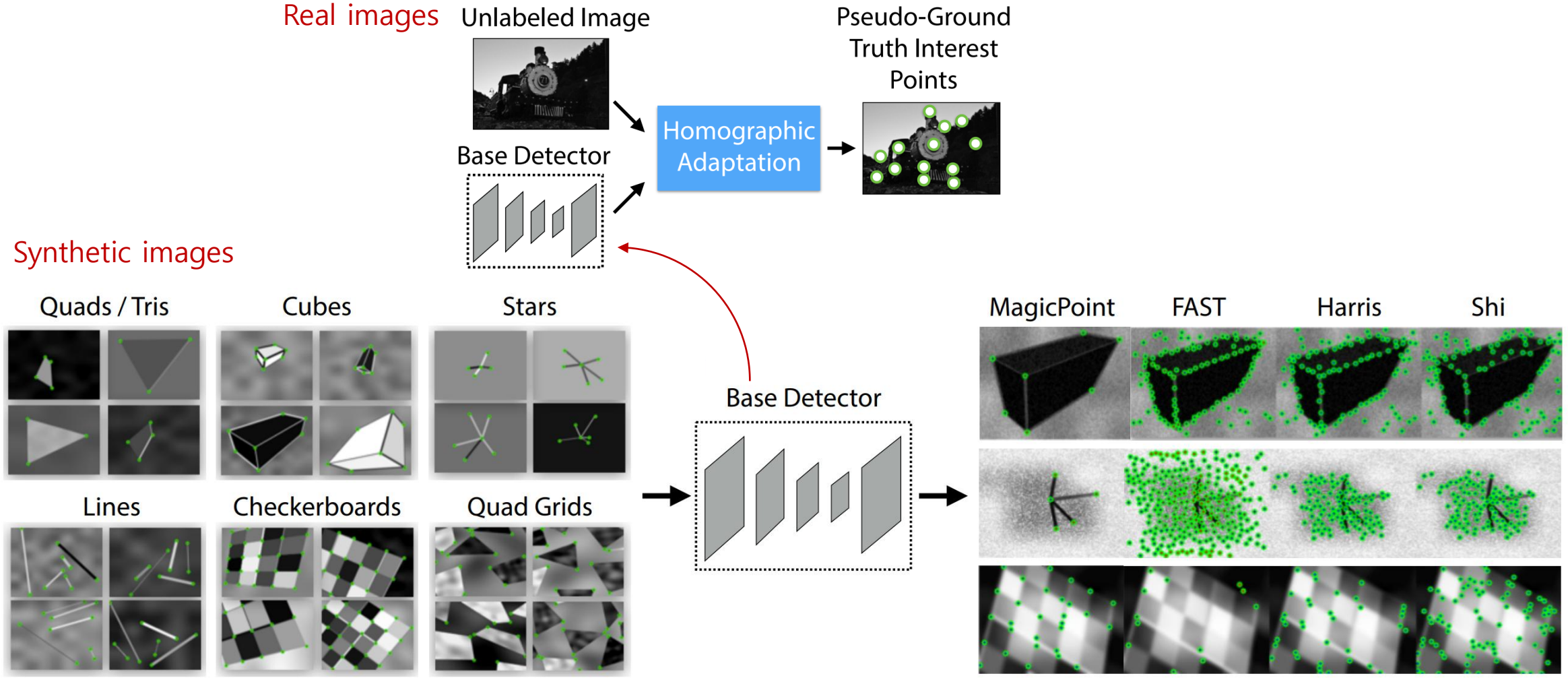


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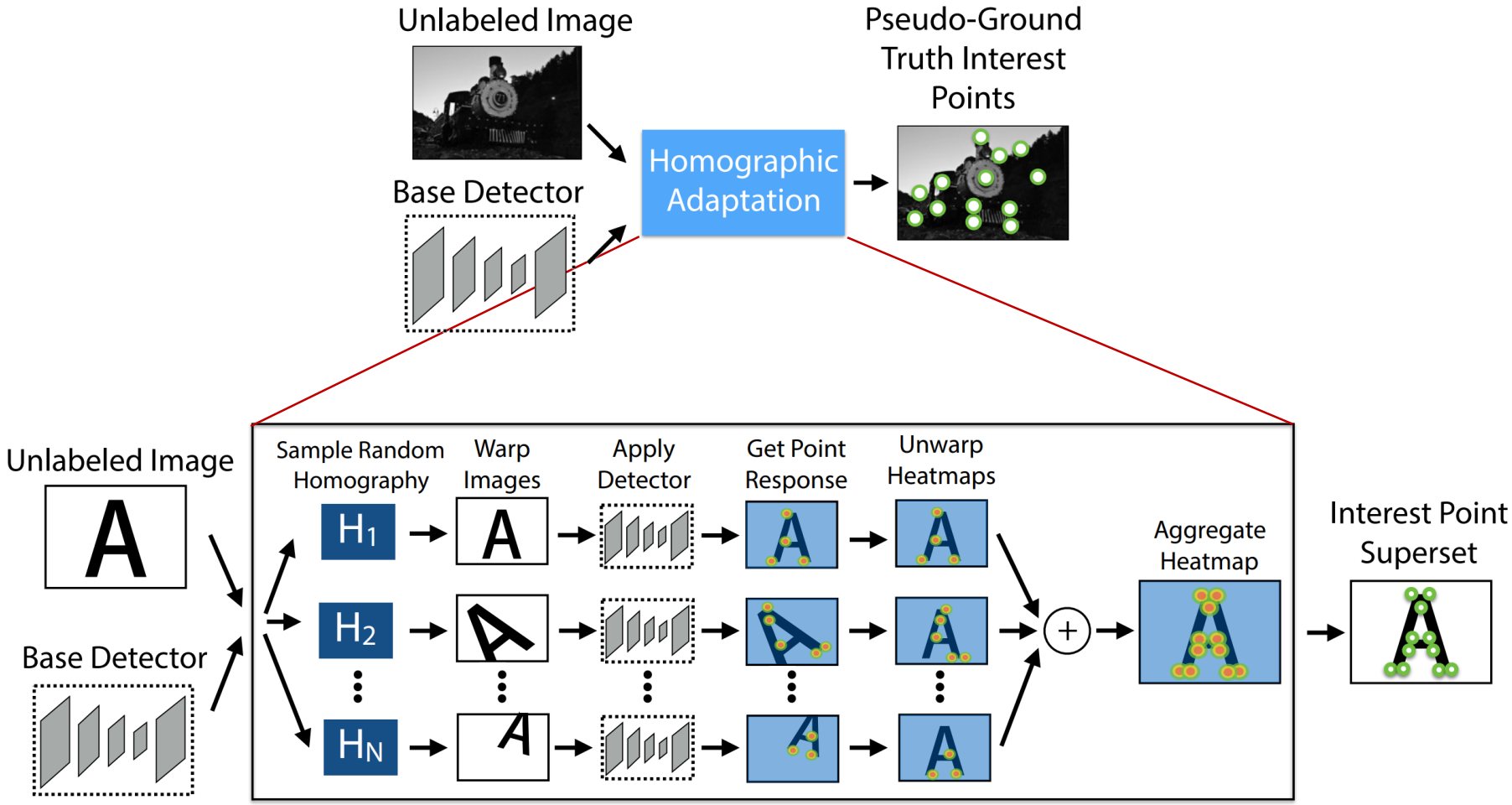


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Feature Point and Descriptor) SuperPoint (2017)

- Key idea: **Self-supervised training with homography transformation**
  - Training data augmentation: Random homography transformation
  - Loss functions: Interest Point Loss (X, Y) + Interest Point Loss (X′, Y′) + Descriptor Loss (Y, Y′)

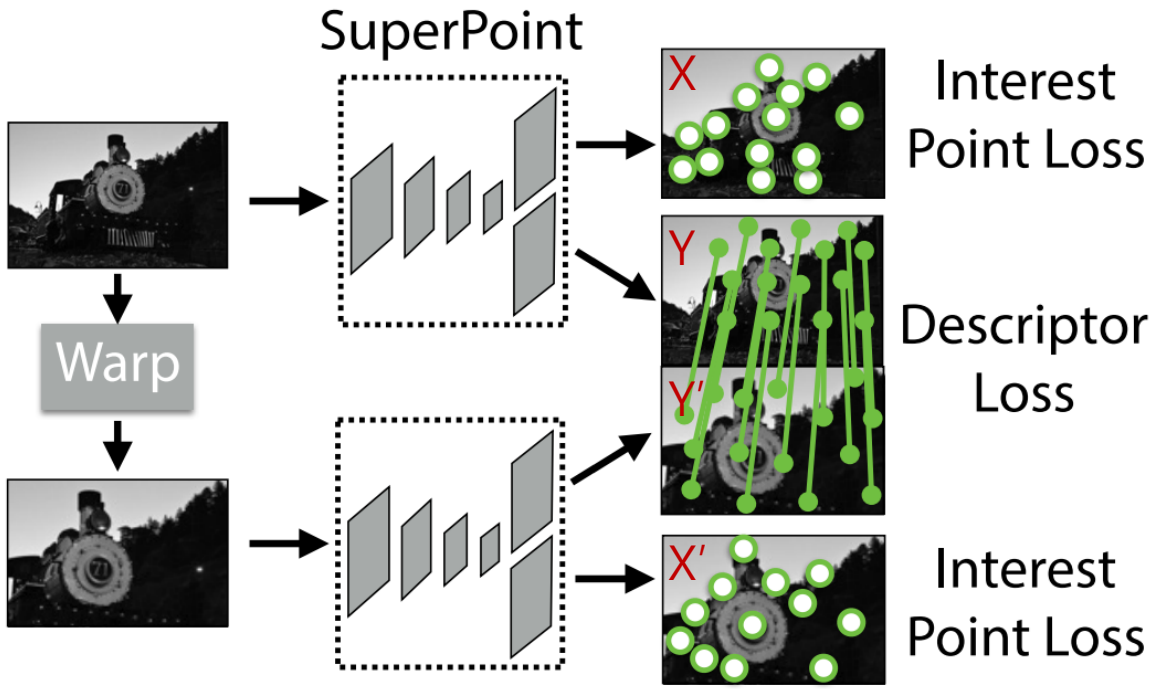
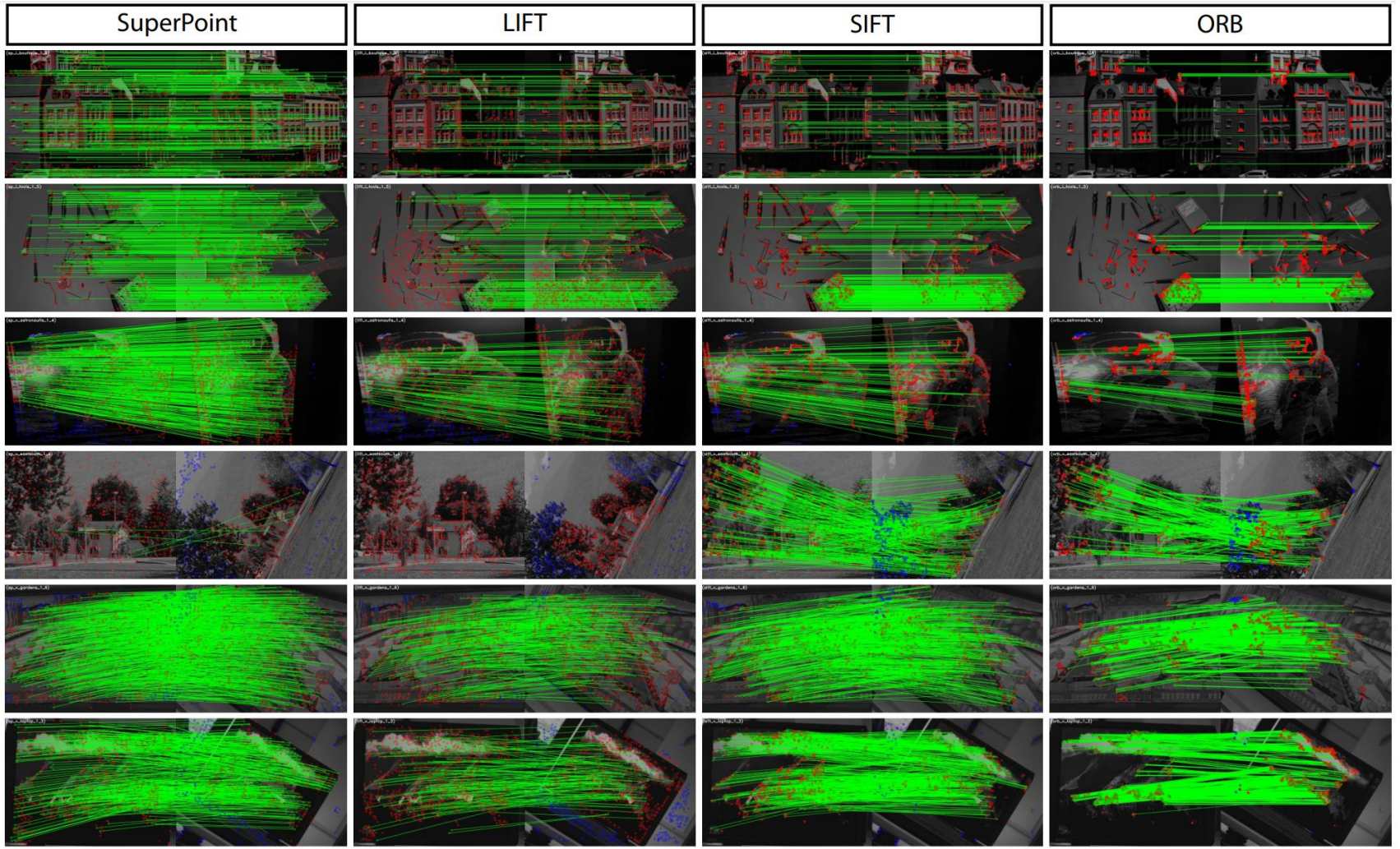


Image: DeTone et al., "SuperPoint: Self-Supervised Interest Point Detection and Description", CVPRW, 2018 arXiv Github (authors) Github

Feature Point and Descriptor) SuperPoint (2017)

- Key idea: **Self-supervised training with homography transformation**
  - Real-time performance: **70 FPS** (13 msec) on 640x480 images with NVIDIA Titan X GPU



Freiburg RGBD:



Microsoft 7 Scenes:



MonoVO:

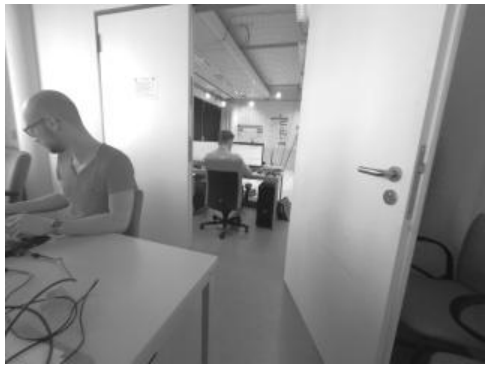
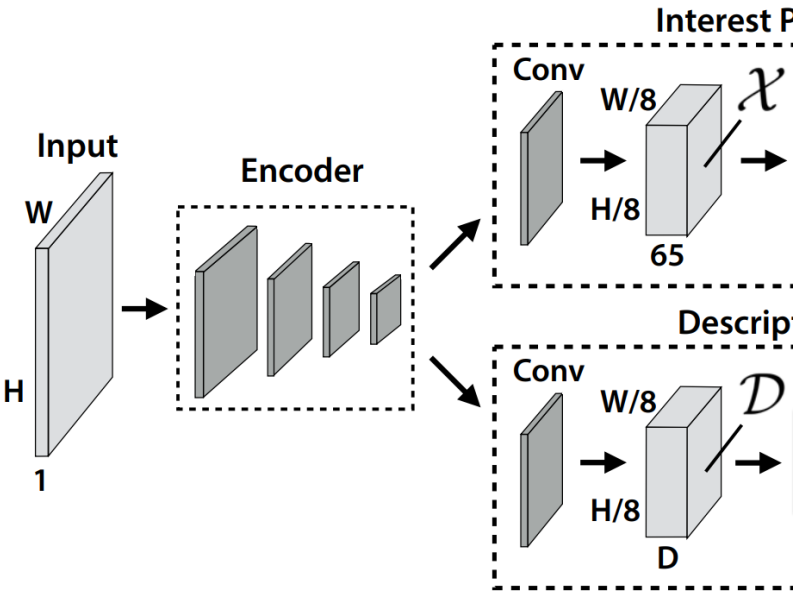


Image: DeTone et al., "SuperPoint: Self-Supervised Interest Point Detection and Description", CVPRW, 2018 arXiv Github (authors) Github

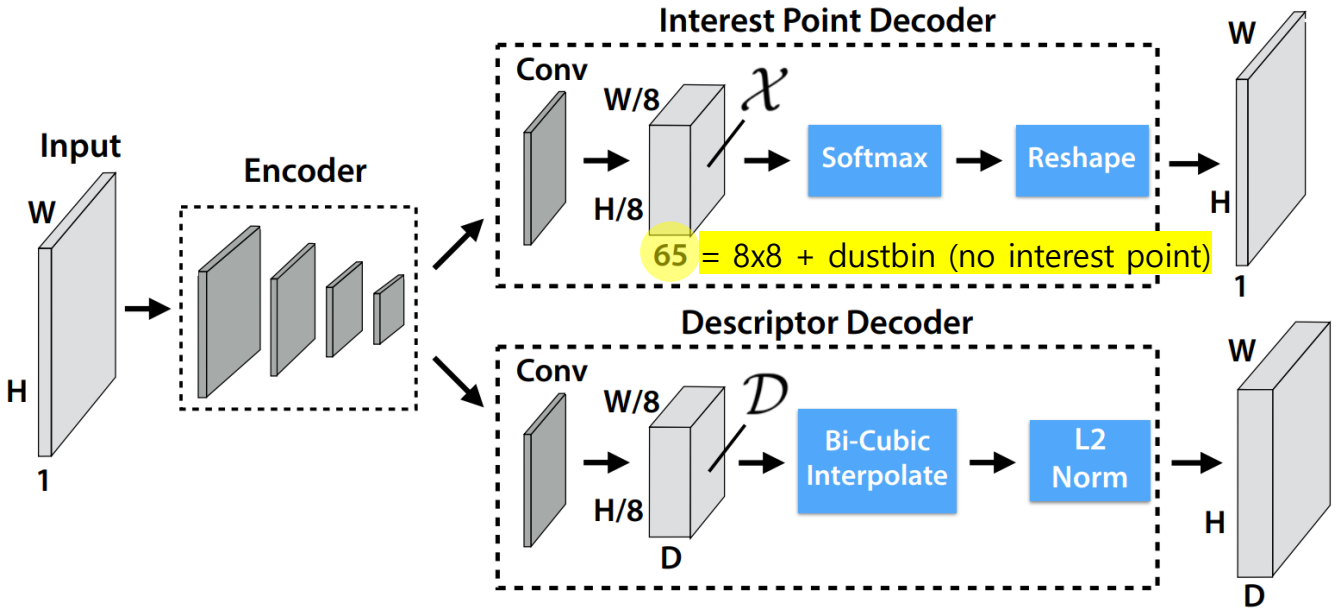
Feature Point and Descriptor) SuperPoint (2017)

- Example) **SuperPoint implementation** [demo_superpoint.py] @ Github (authors)

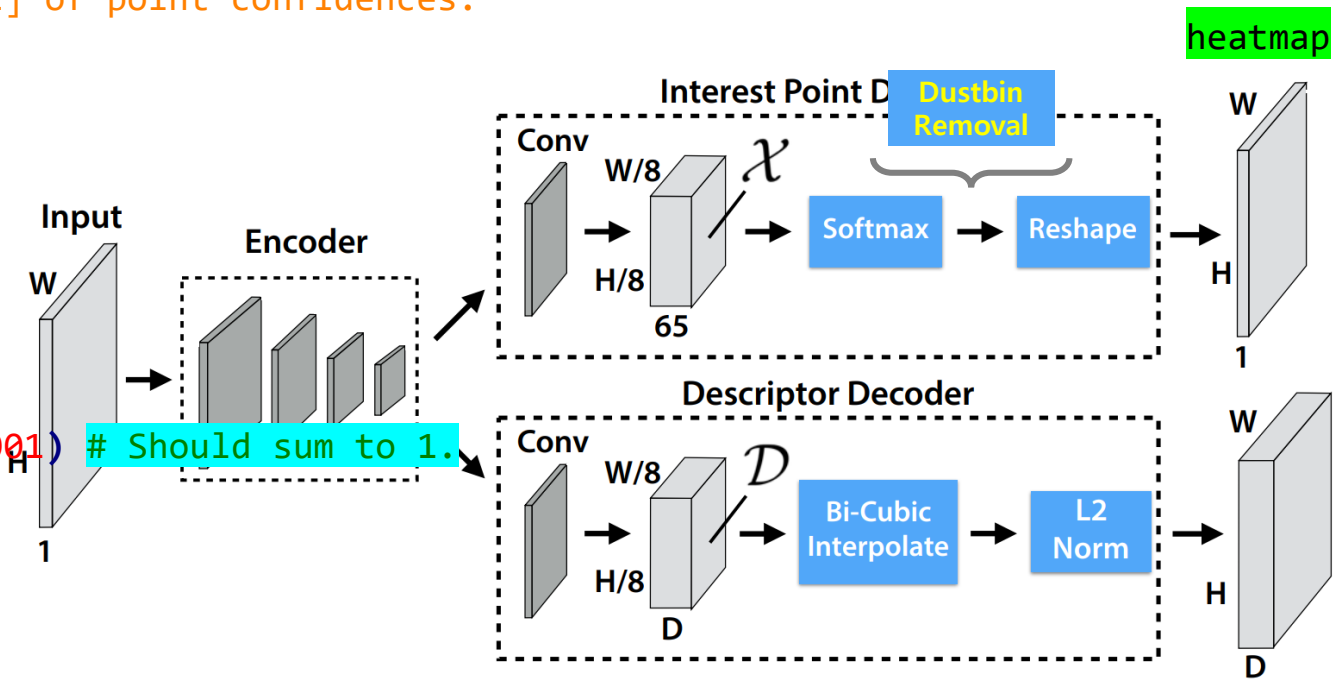
```
class SuperPointNet(torch.nn.Module):
  """ Pytorch definition of SuperPoint Network. """
  def __init__(self):
    super(SuperPointNet, self).__init__()
    self.relu = torch.nn.ReLU(inplace=True)
    self.pool = torch.nn.MaxPool2d(kernel_size=2, stride=2)
    c1, c2, c3, c4, c5, d1 = 64, 64, 128, 128, 256, 256
    # Shared Encoder.
    self.conv1a = torch.nn.Conv2d(1, c1, kernel_size=3, stride=1, padding=1)
    self.conv1b = torch.nn.Conv2d(c1, c1, kernel_size=3, stride=1, padding=1)
    self.conv2a = torch.nn.Conv2d(c1, c2, kernel_size=3, stride=1, padding=1)
    self.conv2b = torch.nn.Conv2d(c2, c2, kernel_size=3, stride=1, padding=1)
    self.conv3a = torch.nn.Conv2d(c2, c3, kernel_size=3, stride=1, padding=1)
    self.conv3b = torch.nn.Conv2d(c3, c3, kernel_size=3, stride=1, padding=1)
    self.conv4a = torch.nn.Conv2d(c3, c4, kernel_size=3, stride=1, padding=1)
    self.conv4b = torch.nn.Conv2d(c4, c4, kernel_size=3, stride=1, padding=1)
    # Detector Head.
    self.convPa = torch.nn.Conv2d(c4, c5, kernel_size=3, stride=1, padding=1)
    self.convPb = torch.nn.Conv2d(c5, 65, kernel_size=1, stride=1, padding=0)
    # Descriptor Head.
    self.convDa = torch.nn.Conv2d(c4, c5, kernel_size=3, stride=1, padding=1)
    self.convDb = torch.nn.Conv2d(c5, d1, kernel_size=1, stride=1, padding=0)
```



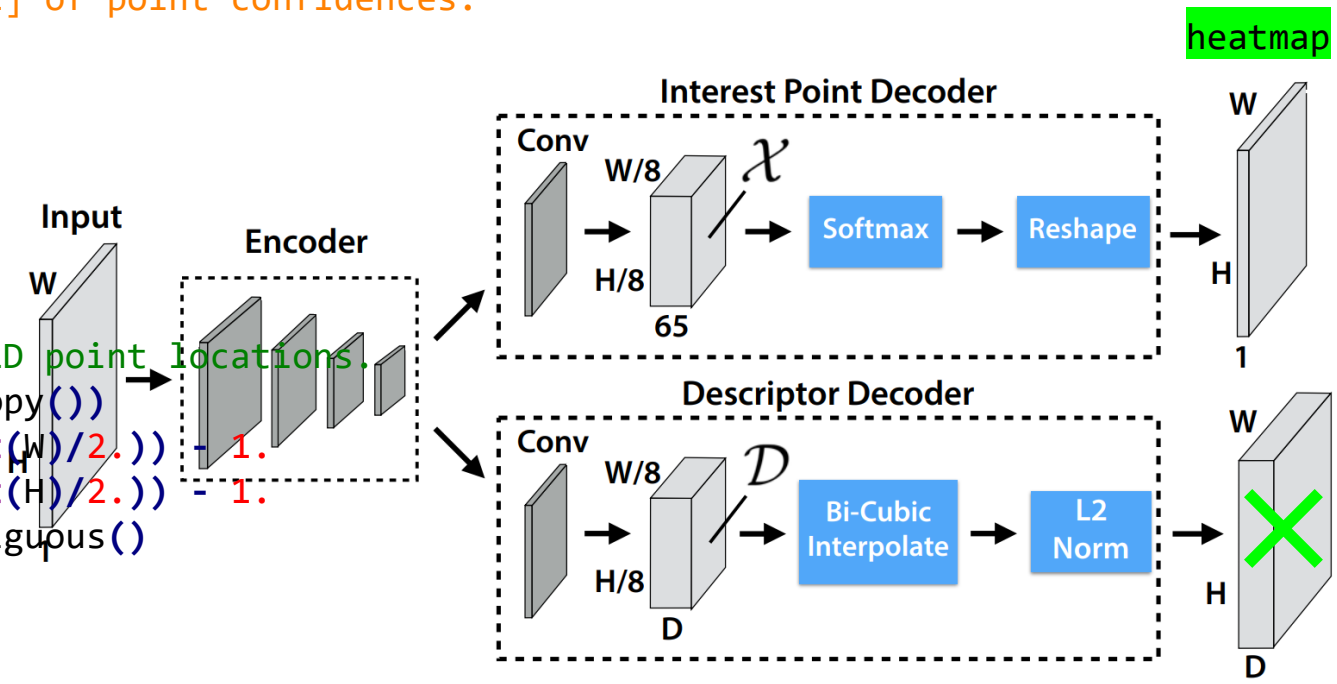
```python
class SuperPointNet(torch.nn.Module):
  ...
  def forward(self, x):
    """ Forward pass that jointly computes unprocessed point and descriptor
    tensors.
    Input
      x: Image pytorch tensor shaped N x 1 x H x W.
    Output
      semi: Output point pytorch tensor shaped N x 65 x H/8 x W/8. (Note: 65 channels includes a dustbin.)
      desc: Output descriptor pytorch tensor shaped N x 256 x H/8 x W/8.
    """
    # Shared Encoder.
    x = self.relu(self.conv1a(x))
    x = self.relu(self.conv1b(x))
    x = self.pool(x)
    x = self.relu(self.conv2a(x))
    x = self.relu(self.conv2b(x))
    x = self.pool(x)
    x = self.relu(self.conv3a(x))
    x = self.relu(self.conv3b(x))
    x = self.pool(x)
    x = self.relu(self.conv4a(x))
    x = self.relu(self.conv4b(x))
    # Detector Head.
    cPa = self.relu(self.convPa(x))
    semi = self.convPb(cPa)
    # Descriptor Head.
    cDa = self.relu(self.convDa(x))
    desc = self.convDb(cDa)
    dn = torch.norm(desc, p=2, dim=1) # Compute the norm.
    desc = desc.div(torch.unsqueeze(dn, 1)) # Divide by norm to normalize.
    return semi, desc
```



```python
class SuperPointNet(torch.nn.Module):
  ...
  def run(self, img):
    """ Process a numpy image to extract points and descriptors.
    Input
      img - HxW numpy float32 input image in range [0,1].
    Output
      corners - 3xN numpy array with corners [x_i, y_i, confidence_i]^T.
      desc - 256xN numpy array of corresponding unit normalized descriptors.
      heatmap - HxW numpy heatmap in range [0,1] of point confidences.
    """
    ...
    # Forward pass of network.
    outs = self.net.forward(inp)
    semi, coarse_desc = outs[0], outs[1]
    # Convert pytorch -> numpy.
    semi = semi.data.cpu().numpy().squeeze()
    # --- Process points.
    dense = np.exp(semi) # Softmax.
    dense = dense / (np.sum(dense, axis=0)+.00001) # Should sum to 1.
    # Remove dustbin.
    nodust = dense[:-1, :, :]
    # Reshape to get full resolution heatmap.
    Hc = int(H / self.cell)
    Wc = int(W / self.cell)
    nodust = nodust.transpose(1, 2, 0)
    heatmap = np.reshape(nodust, [Hc, Wc, self.cell, self.cell])
    heatmap = np.transpose(heatmap, [0, 2, 1, 3])
    heatmap = np.reshape(heatmap, [Hc*self.cell, Wc*self.cell])
    ...
```



```python
class SuperPointNet(torch.nn.Module):
  ...
  def run(self, img):
    """ Process a numpy image to extract points and descriptors.
    Input
      img - HxW numpy float32 input image in range [0,1].
    Output
      corners - 3xN numpy array with corners [x_i, y_i, confidence_i]^T.
      desc - 256xN numpy array of corresponding unit normalized descriptors.
      heatmap - HxW numpy heatmap in range [0,1] of point confidences.
    """
    ...
    # --- Process descriptor.
    D = coarse_desc.shape[1]
    if pts.shape[1] == 0:
      desc = np.zeros((D, 0))
    else:
      # Interpolate into descriptor map using 2D point locations.
      samp_pts = torch.from_numpy(pts[:2, :].copy())
      samp_pts[0, :] = (samp_pts[0, :] / (float(W)/2.)) - 1.
      samp_pts[1, :] = (samp_pts[1, :] / (float(H)/2.)) - 1.
      samp_pts = samp_pts.transpose(0, 1).contiguous()
      samp_pts = samp_pts.view(1, 1, -1, 2)
      samp_pts = samp_pts.float()
      if self.cuda:
        samp_pts = samp_pts.cuda()
      desc = torch.nn.functional.grid_sample(coarse_desc, samp_pts)
      desc = desc.data.cpu().numpy().reshape(D, -1)
      desc /= np.linalg.norm(desc, axis=0)[np.newaxis, :]
    return pts, desc, heatmap
```



Summary) Feature Points and Descriptors

Feature Points	**Gradient-based** ▪ Harris ▪ GFTT (a.k.a. Shi-Tomasi) ▪ SIFT ▪ SURF	**Intensity-based** ▪ FAST	**DL-based** ▪ LIFT ▪ SuperPoint
Feature Descriptor	**Real-valued** ▪ SIFT ▪ SURF	**Binary-valued** ▪ BRIEF ▪ ORB (FAST+BRIEF)	**Real-valued (DL-based)** ▪ LIFT ▪ SuperPoint
Advantages Disadvantages	(+) Accurate (–) Slow	(+) Fast (–) Inaccurate (+) Less storage	(+) Accurate (+) Fast (–) GPU requirement

Image: pixabay

Table of Contents

- **Feature Points**
  - Gradient-based: Harris corner, GFTT corner, SIFT, SURF
  - Intensity-based: FAST
  - DL-based: LIFT, SuperPoint
- **Feature Descriptors**
  - Real-valued: SIFT, SURF (DL-based: LIFT, SuperPoint)
  - Binary-valued: BRIEF, ORB
- **Feature Matching**
  - Q) How can we associate the points across different images?
- **Feature Tracking**
  - Q) How can we associate the points across their next image?
- **Outlier Rejection**
  - Q) How can we select correctly matched points?



Feature Matching) For Real-valued Descriptors

- **Real-valued descriptors**
  - Distance measures
    - Euclidean distance: $l_2(\mathbf{d}, \mathbf{d}') = \|\mathbf{d} - \mathbf{d}'\|_2$ ($\downarrow$: similar)
    - Cosine similarity: $s_c(\mathbf{d}, \mathbf{d}') = \frac{\mathbf{d} \cdot \mathbf{d}'}{\|\mathbf{d}\| \|\mathbf{d}'\|}$ ($\uparrow$: similar)
    - Note) Matching measures can be combined or advanced.
      - e.g. The ratio of the best and second best similarity > threshold
        - This can select more distinctive matches.
  - Matching algorithms
    - Brute-force search
      - Time complexity: $O(N)$ for N descriptors
    - Approximate nearest neighborhood search (ANN search)
      - Time complexity: $O(\log N)$ or less for N descriptors
      - Note) Big-ANN Competition (recent: NeurIPS 2023)

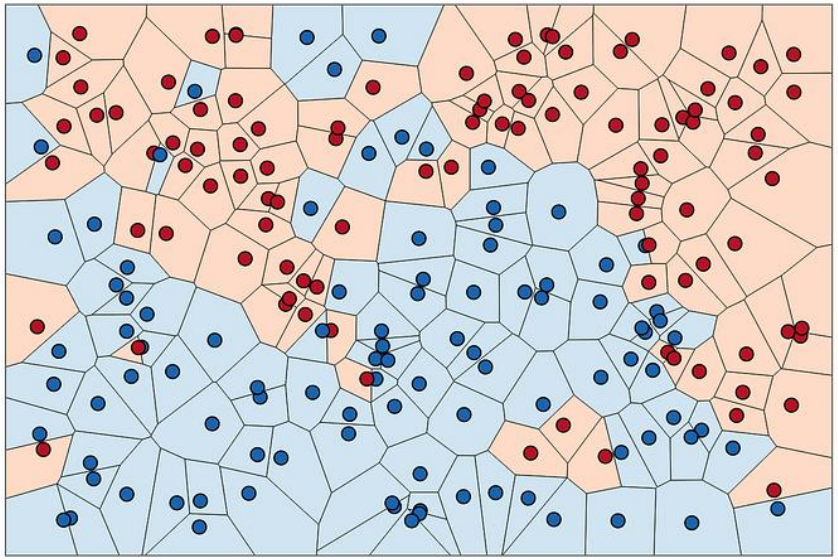


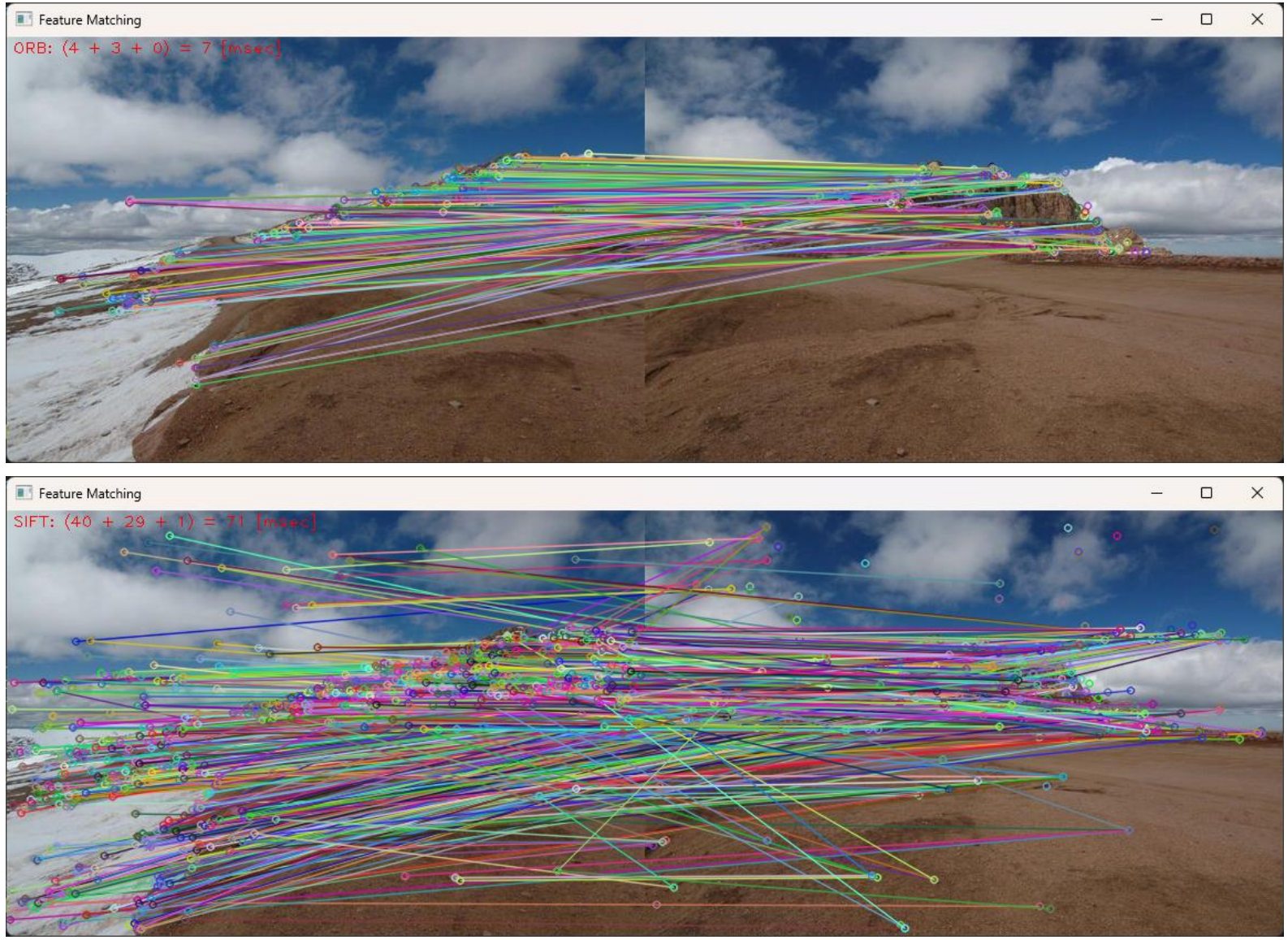
Image: George Williams, "Big-ANN-Benchmarks", Medium

Feature Matching) For Binary-valued Descriptors

- **Binary-valued descriptors**
  - Distance measures
    - Hamming distance: $l_h(\mathbf{d}, \mathbf{d}') = \sum_i (d_i \neq d_i')$
      - e.g. 4-bit descriptors
        - $l_h(0110, 1110) = 1$ vs. (6, 14)
        - $l_h(0110, 0111) = 1$ vs. (6, 7)
        - $l_h(0110, 0101) = 2$ vs. (6, 5)
      - Note) Hamming distance is the L_1-norm with binary-valued descriptors.
  - Matching algorithms
    - Brute-force search
      - Time complexity: $O(N)$ for N descriptors

Feature Matching

- Example) **Feature matching comparison** `[feature_matching.py]`



Feature Matching

- Example) **Feature matching comparison** [feature_matching.py]

```
# Load two images
img1 = cv.imread('../data/hill01.jpg')
img2 = cv.imread('../data/hill02.jpg')

# Instantiate feature detectors and matchers
# Note) You can specify options for each detector in its creation.
features = [
    ...
    {'name': 'FAST',    'detector': cv.FastFeatureDetector_create(),
                        'matcher' : None}, # No descriptor
    ...
    {'name': 'ORB',     'detector': cv.ORB_create(),
                        'matcher' : cv.DescriptorMatcher_create('BruteForce-Hamming')},
    {'name': 'SIFT',    'detector': cv.SIFT_create(),
                        'matcher' : cv.DescriptorMatcher_create('BruteForce')},
]
```

Feature Matching

- Example) **Feature matching comparison** [feature_matching.py]

```
# Detect feature points
keypoints1 = features[f_select]['detector'].detect(img1)
keypoints2 = features[f_select]['detector'].detect(img2)

if features[f_select]['matcher'] is not None:
    # Extract feature descriptors
    keypoints1, descriptors1 = features[f_select]['detector'].compute(img1, keypoints1)
    keypoints2, descriptors2 = features[f_select]['detector'].compute(img2, keypoints2)

    # Match the feature descriptors
    match = features[f_select]['matcher'].match(descriptors1, descriptors2)

# Show the matched image
if features[f_select]['matcher'] is not None:
    img_merged = cv.drawMatches(img1, keypoints1, img2, keypoints2, match, None)
else:
    img1_keypts = cv.drawKeypoints(img1, keypoints1, None)
    img2_keypts = cv.drawKeypoints(img2, keypoints2, None)
    img_merged = np.hstack((img1_keypts, img2_keypts))
info = features[f_select]['name’] + ...
cv.putText(img_merged, info, (5, 15), cv.FONT_HERSHEY_PLAIN, 1, (0, 0, 255))
cv.imshow('Feature Matching', img_merged)
```

Feature Matching) SuperGlue (2020) → LightGlue (2023)

- Key idea: **Encoding feature points and descriptors using a graph neural network with attention**
  - For extreme wide-baseline image pairs in real-time on GPU
  - e.g. Relative pose estimation on the ScanNet dataset (more correct matches and less mismatches)

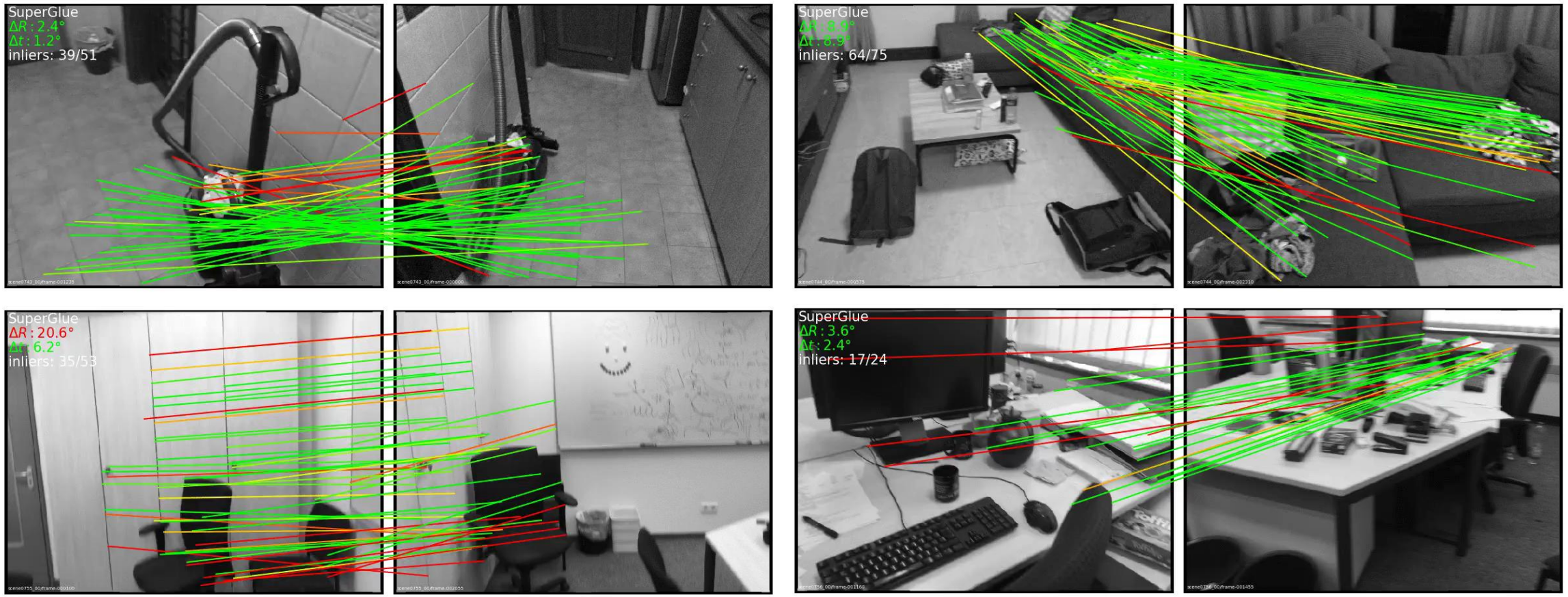
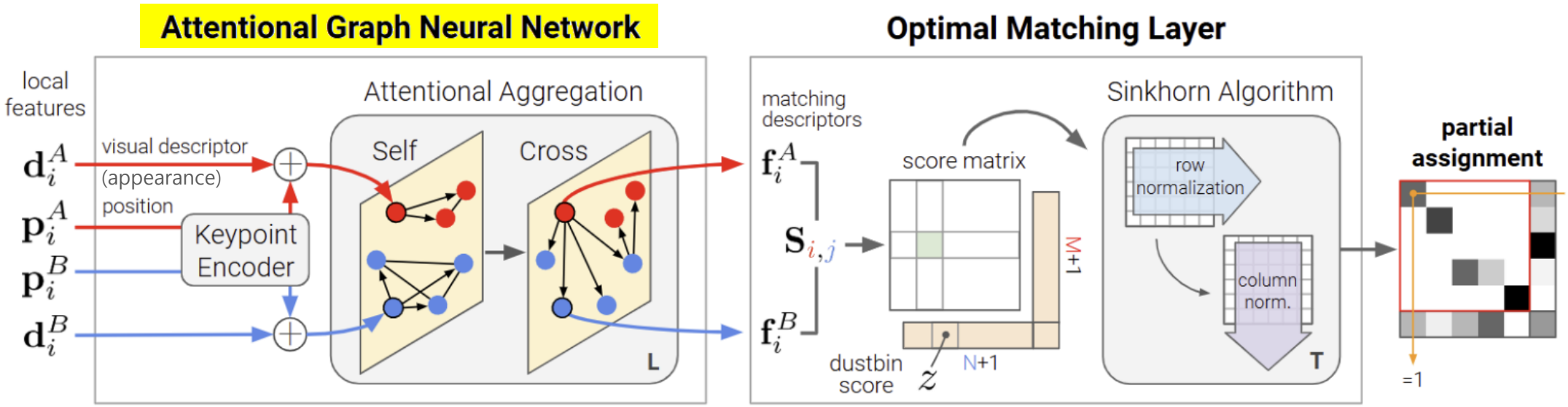


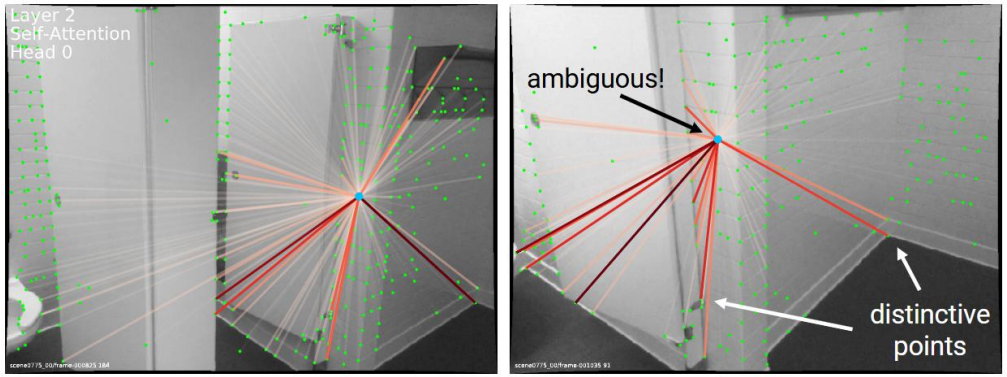
Image: Sarlin et al., "SuperGlue: Learning Feature Matching with Graph Neural Networks", CVPR, 2020 arXiv Github (authors) Github Homepage

Feature Matching) SuperGlue (2020)

- Key idea: **Encoding feature points and descriptors using a graph neural network with attention**



Self-attention: Intra-image information flow



Cross-attention: Inter-image communication

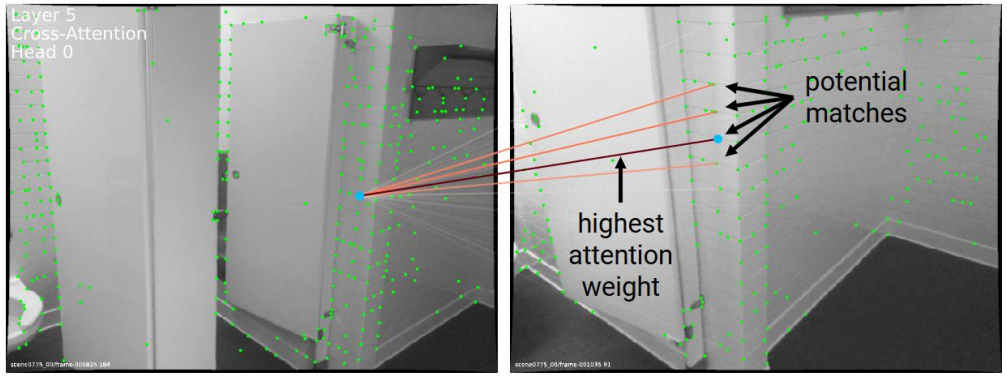
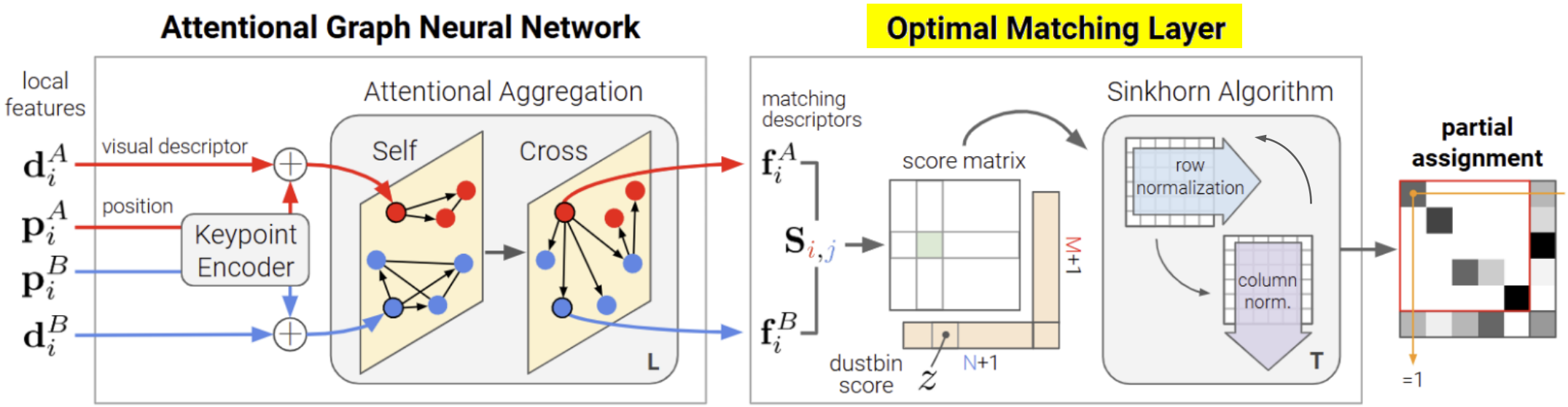


Image: Sarlin et al., "SuperGlue: Learning Feature Matching with Graph Neural Networks", CVPR, 2020 arXiv Github (authors) Github Homepage

Feature Matching) SuperGlue (2020)

- Key idea: **Encoding feature points and descriptors using a graph neural network with attention**



- Matching score $\mathbf{S}_{i,j}$: Inner product of $\mathbf{f}_i^A$ and $\mathbf{f}_j^B$
- Sinkhorn algorithm: Differentiable & soft [Hungarian algorithm]

e.g. Optimal assignment problem

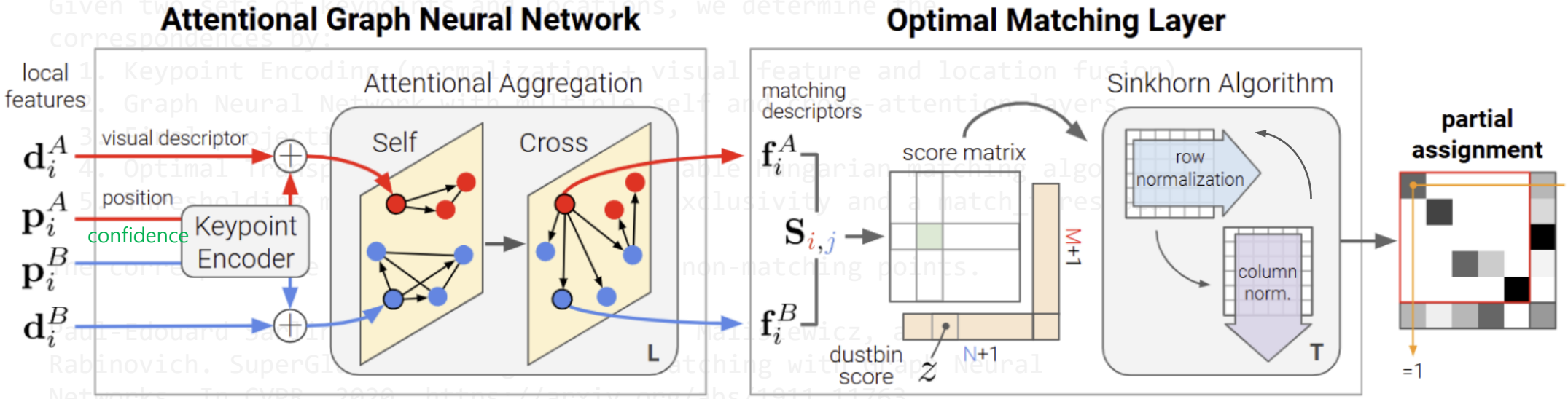
Worker \ Task	Clean bathroom	Sweep floors	Wash windows
Alice	**$8**	$4	$7
Bob	$5	$2	**$3**
Carol	$9	**$4**	$8

Image: Sarlin et al., "SuperGlue: Learning Feature Matching with Graph Neural Networks", CVPR, 2020 arXiv Github (authors) Github Homepage

Feature Matching) SuperGlue (2020)

- Example) **SuperGlue implementation** [models/superglue.py] @ Github (authors)

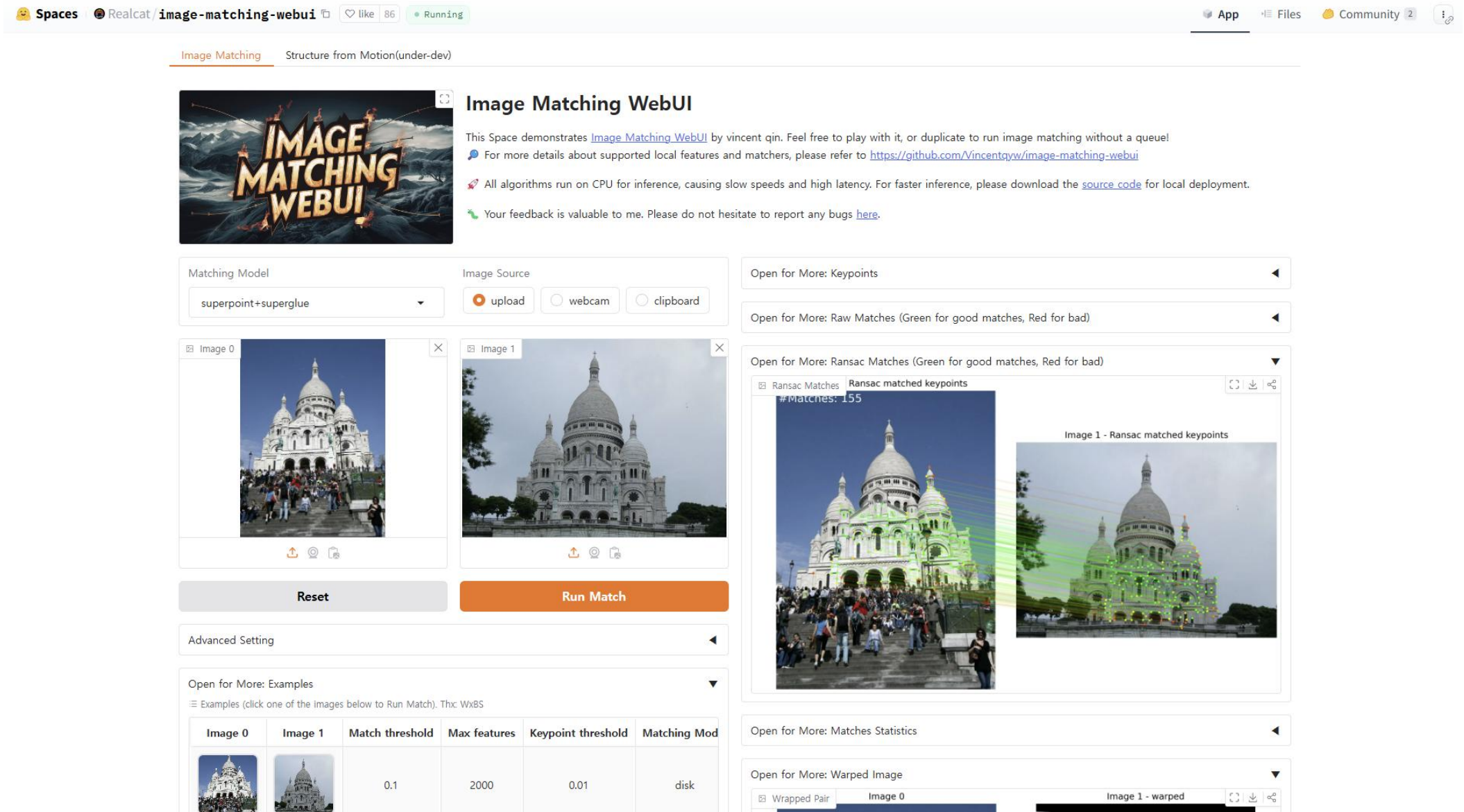
```
class SuperGlue(nn.Module):
    """SuperGlue feature matching middle-end
```




```
    """
    default_config = {
        'descriptor_dim': 256,
        'weights': 'indoor',
        'keypoint_encoder': [32, 64, 128, 256], # 4-layer CNN: [3, 32, 128, 256, 256] channels (kernel size = 1)
        'GNN_layers': ['self', 'cross'] * 9,    # 18-layer attention network + 1-layer CNN (final projection)
        'sinkhorn_iterations': 100,             # T = 100
        'match_threshold': 0.2,
    }
```

Feature Extraction and Matching

- Example) Feature matching comparison: [Image Matching WebUI]



~~Feature~~ Template Matching) For Image Patches

- **Raw image patches** (or **histogram**) can be used as descriptors.
  - Distance measures
    - SAD (sum of absolute difference): $l_1(\mathbf{d}, \mathbf{d}') = \|\mathbf{d} - \mathbf{d}'\|_1 = \sum_i |d_i - d'_i|$
    - SSD (sum of squared difference): $l_2^2(\mathbf{d}, \mathbf{d}') = \|\mathbf{d} - \mathbf{d}'\|_2^2 = \sum_i (d_i - d'_i)^2$
    - Cross-correlation: $s_{cc}(\mathbf{d}, \mathbf{d}') = \mathbf{d} \cdot \mathbf{d}' = \sum_i d_i d'_i$
    - NCC (normalized cross-correlation): $s_{NCC}(\mathbf{d}, \mathbf{d}') = \frac{\mathbf{d} \cdot \mathbf{d}'}{\|\mathbf{d}\| \|\mathbf{d}'\|}$ (~ cosine similarity)
    - ZNCC (zero-mean normalized cross-correlation): $s_{ZNCC}(\mathbf{d}, \mathbf{d}') = \frac{\bar{\mathbf{d}} \cdot \bar{\mathbf{d}}'}{\|\bar{\mathbf{d}}\| \|\bar{\mathbf{d}}'\|}$ where $\bar{\mathbf{d}} = \mathbf{d} - \mathrm{E}(\mathbf{d})$ and $\bar{\mathbf{d}}' = \mathbf{d}' - \mathrm{E}(\mathbf{d}')$
  - Matching algorithms
    - Sliding window (**d**: )

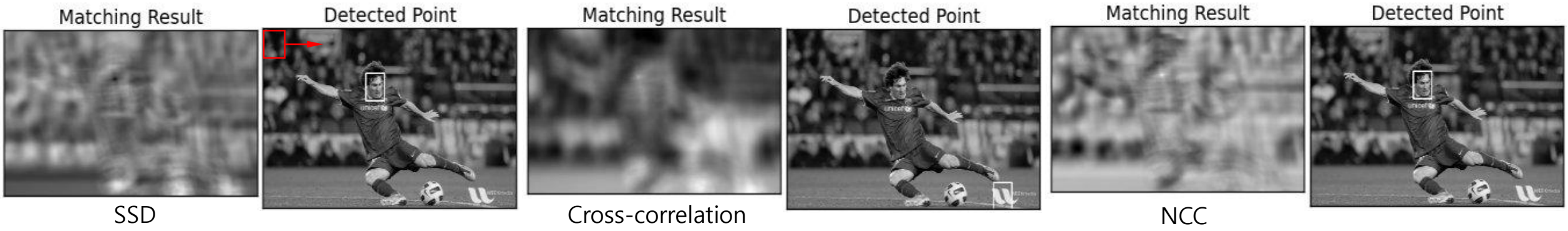


Image: OpenCV Tutorials

Feature Tracking) Lukas-Kanade Optical Flow (1981)

- Key idea: **Finding movement of a patch whose pixel values are the same**
  - Brightness constancy constraint: $I(x, y, t) = I(x + \Delta_x, y + \Delta_y, t + \Delta_t)$

$$I_x \frac{\Delta_x}{\Delta_t} + I_y \frac{\Delta_y}{\Delta_t} + I_t = 0 \quad \leftarrow \quad I(x + \Delta_x, y + \Delta_y, t + \Delta_t) \approx I(x, y, t) + I_x \Delta_x + I_y \Delta_y + I_t \Delta_t$$

$$\mathbf{A}\mathbf{x} = \mathbf{b} \quad \text{where} \quad \mathrm{A} = [I_x \quad I_y],\ \mathbf{x} = \left[\Delta_x,\ \Delta_y\right]^{\mathsf{T}},\ \text{and}\ \mathbf{b} = [-I_t] \quad (\Delta_t = 1)$$

$$\therefore \mathbf{x} = \mathrm{A}^{\dagger}\,\mathbf{b}$$

- Combination: **KLT tracker**
  - GFTT (a.k.a. Shi-**T**omasi) detector + **L**ukas-**K**anade optical flow

- Advantages and disadvantages (feature tracking vs. matching)
  - (+) No descriptor required (→ fast and compact)
  - (—) Continuous feature tracking causes drift errors.
  - (—) Does not work well for wide-baseline cases
  - (+) Able to control the matching range

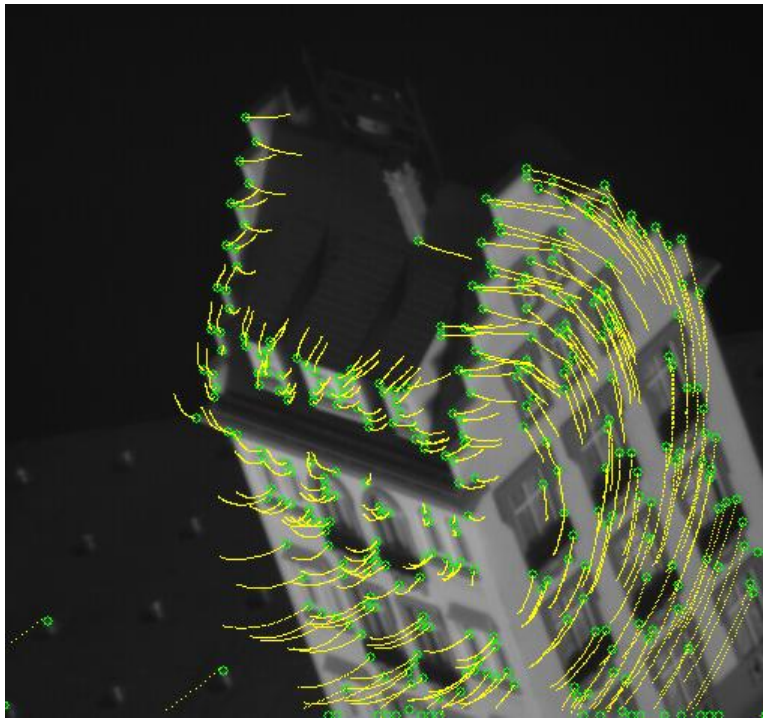


Image: Nirmal Amirthalingam, Github

Feature Tracking) **Lukas-Kanade Optical Flow (1981)**

- Example) **Feature matching comparison** `[feature_tracking_klt.py]`



Feature Tracking) Lukas-Kanade Optical Flow (1981)

- Example) **Feature matching comparison** [feature_tracking_klt.py]

```
# Open a video and get an initial image
video = cv.VideoCapture(video_file)
assert video.isOpened()

_, gray_prev = video.read()
if gray_prev.ndim >= 3 and gray_prev.shape[2] > 1:
    gray_prev = cv.cvtColor(gray_prev, cv.COLOR_BGR2GRAY)

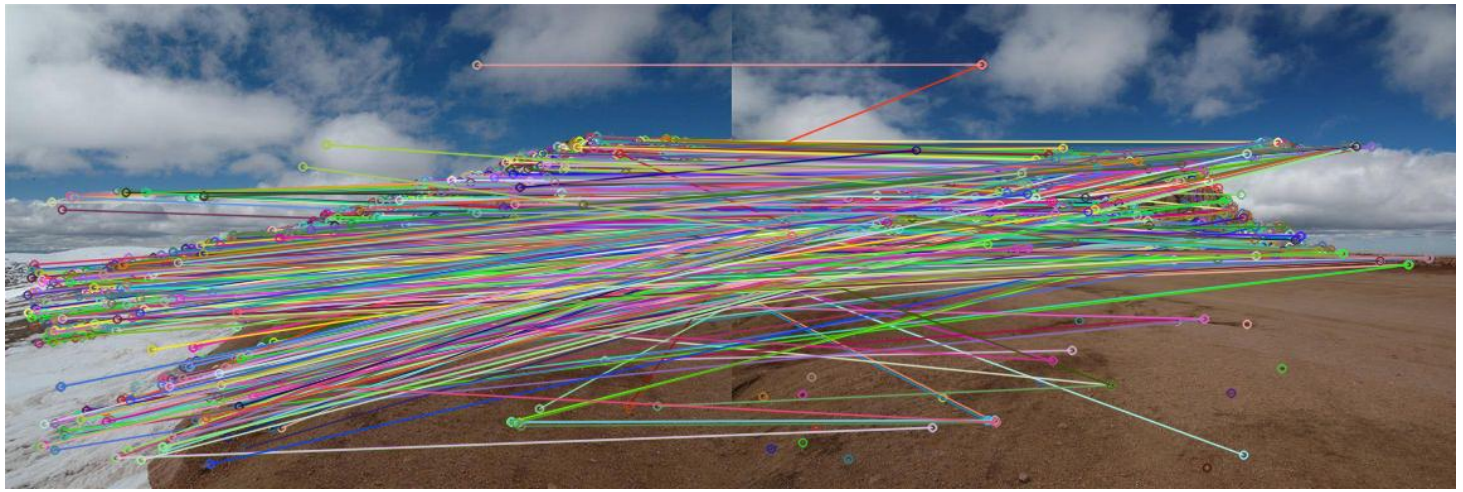
# Run the KLT feature tracker
while True:
    # Grab an image from the video
    valid, img = video.read()
    if not valid:
        break
    if img.ndim >= 3 and img.shape[2] > 1:
        gray = cv.cvtColor(img, cv.COLOR_BGR2GRAY)
    else:
        gray = img.copy()

    # Extract optical flow
    pts_prev = cv.goodFeaturesToTrack(gray_prev, 2000, 0.01, 10)
    pts, status, error = cv.calcOpticalFlowPyrLK(gray_prev, gray, pts_prev, None)
    gray_prev = gray

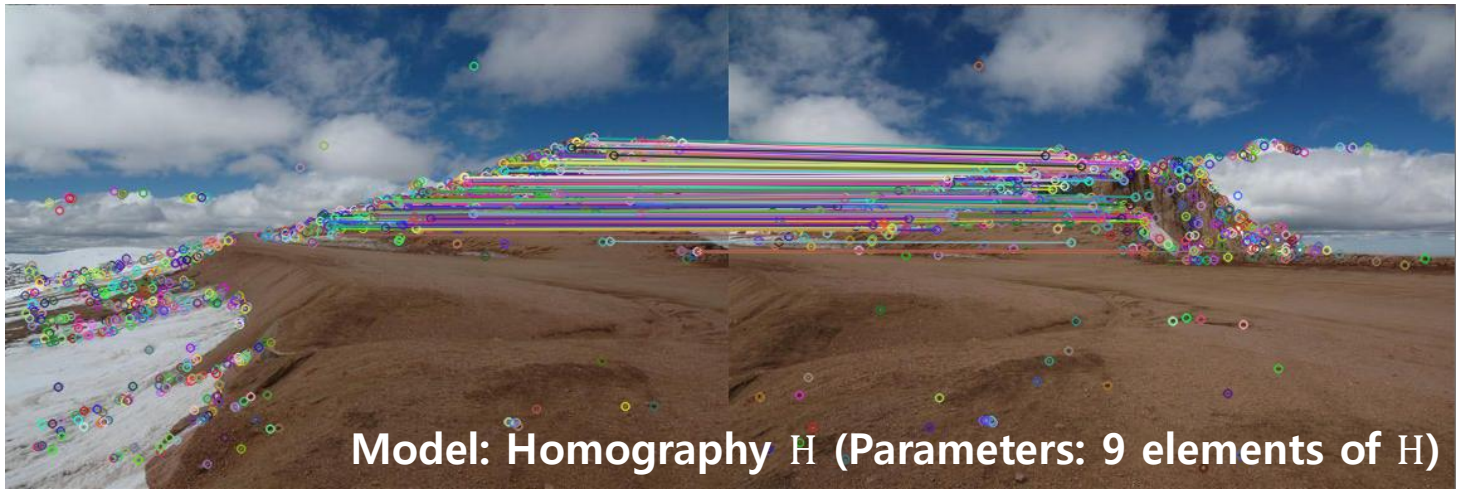
    # Show the optical flow on the image
    if img.ndim < 3 or img.shape[2] < 3:
```

Why Outliers?

Putative feature matches (inliers + outliers)

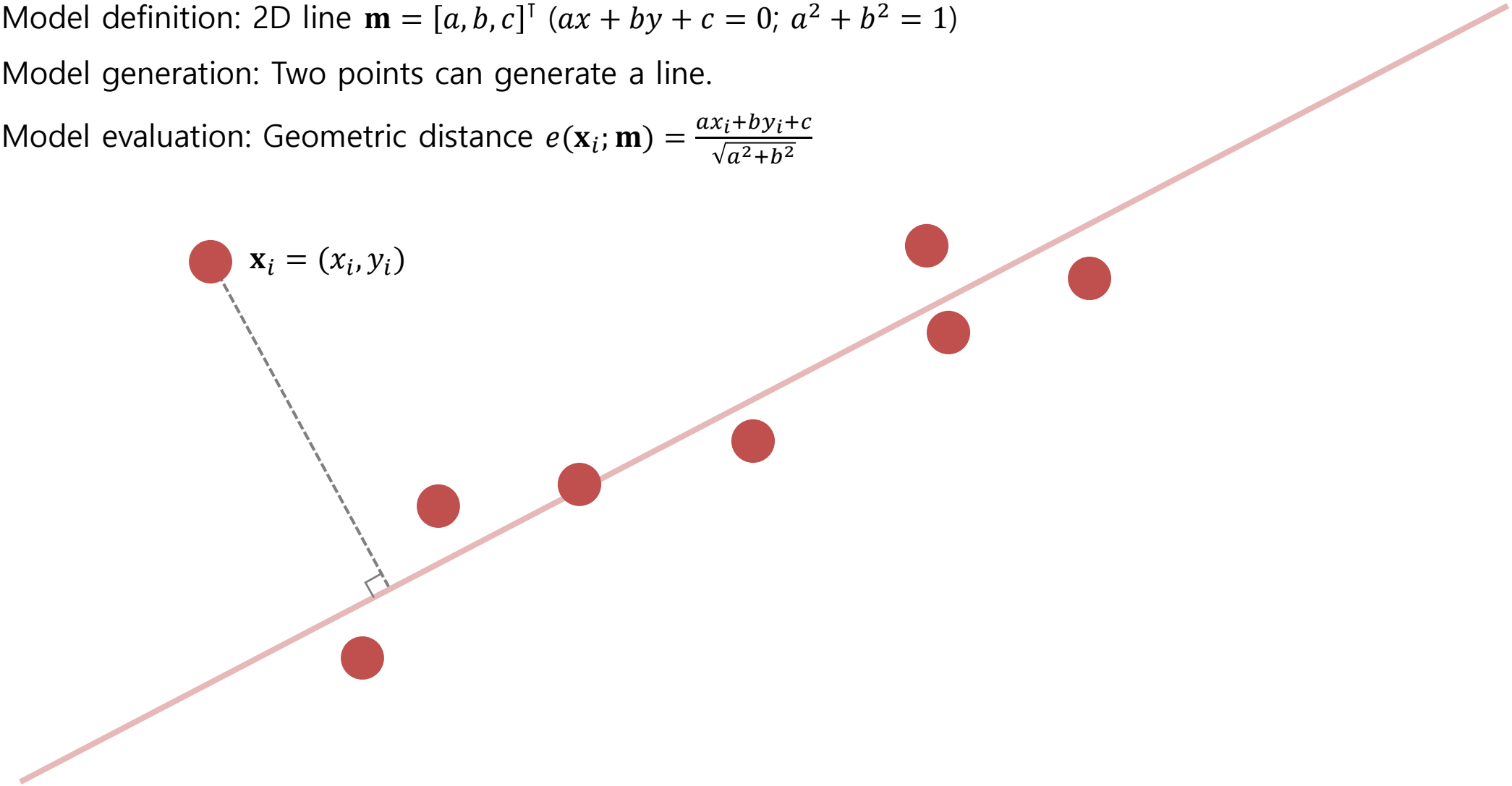


After applying RANSAC (inliers)

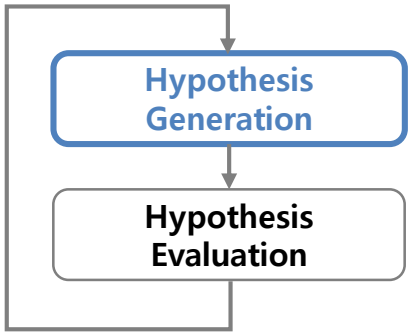


Outlier Rejection) RANSAC (Random Sample Consensus; 1981)

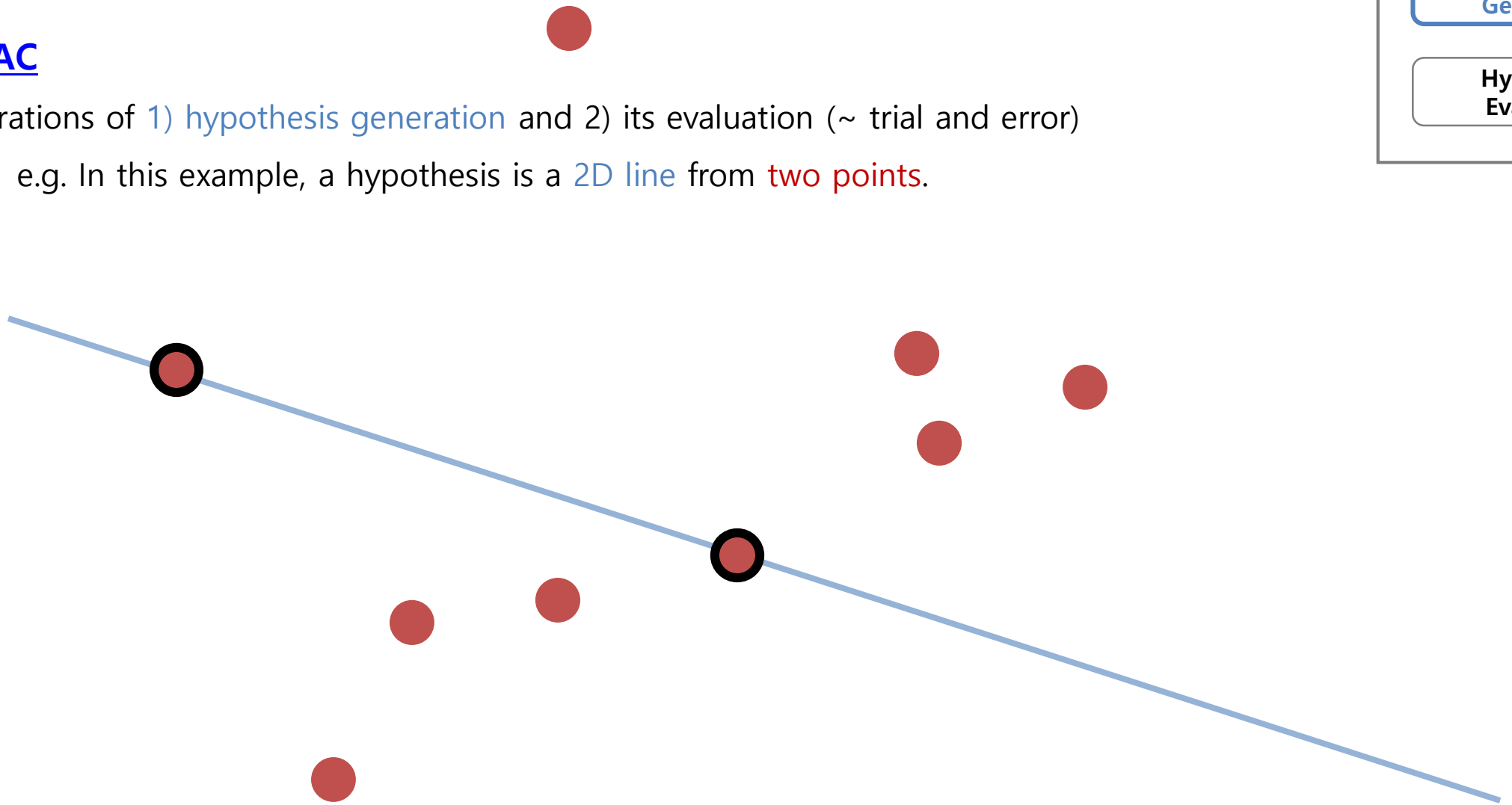
- Example) **Line fitting with RANSAC**
  - Model definition: 2D line $\mathbf{m} = [a, b, c]^{\mathrm{T}}$ $(ax + by + c = 0;\ a^2 + b^2 = 1)$
  - Model generation: Two points can generate a line.
  - Model evaluation: Geometric distance $e(\mathbf{x}_i; \mathbf{m}) = \frac{ax_i + by_i + c}{\sqrt{a^2 + b^2}}$



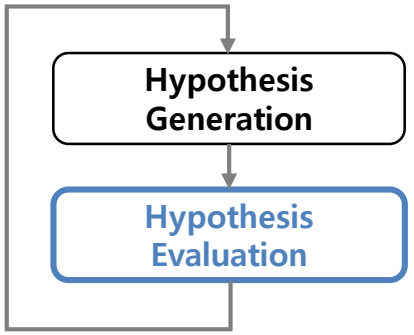
Outlier Rejection) RANSAC (Random Sample Consensus; 1981)



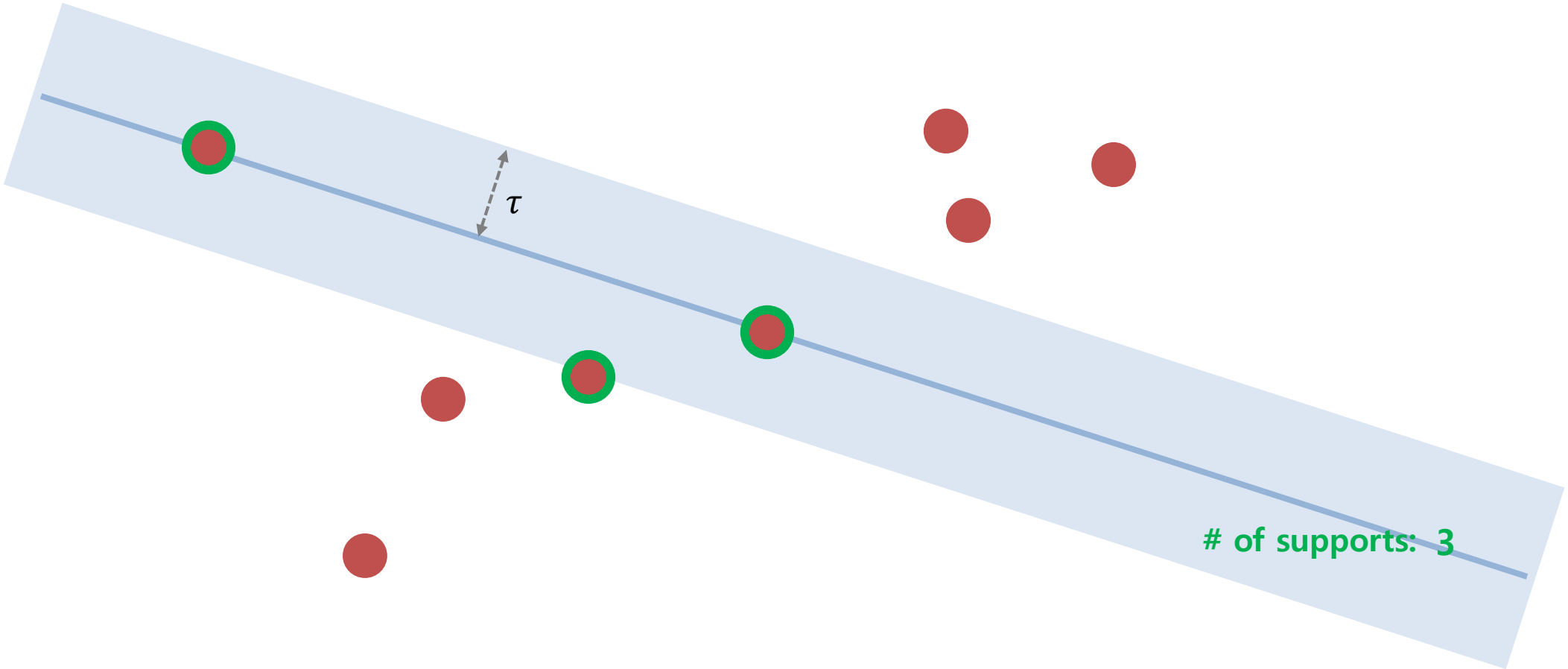
- **RANSAC**
  - Iterations of 1) hypothesis generation and 2) its evaluation (~ trial and error)
    - e.g. In this example, a hypothesis is a 2D line from two points.



Outlier Rejection) RANSAC (Random Sample Consensus; 1981)

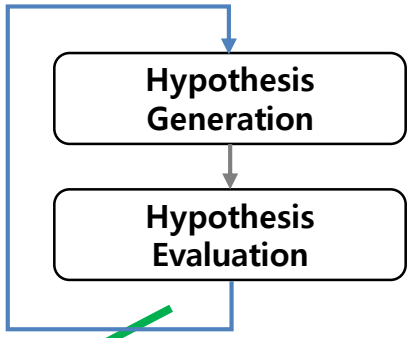


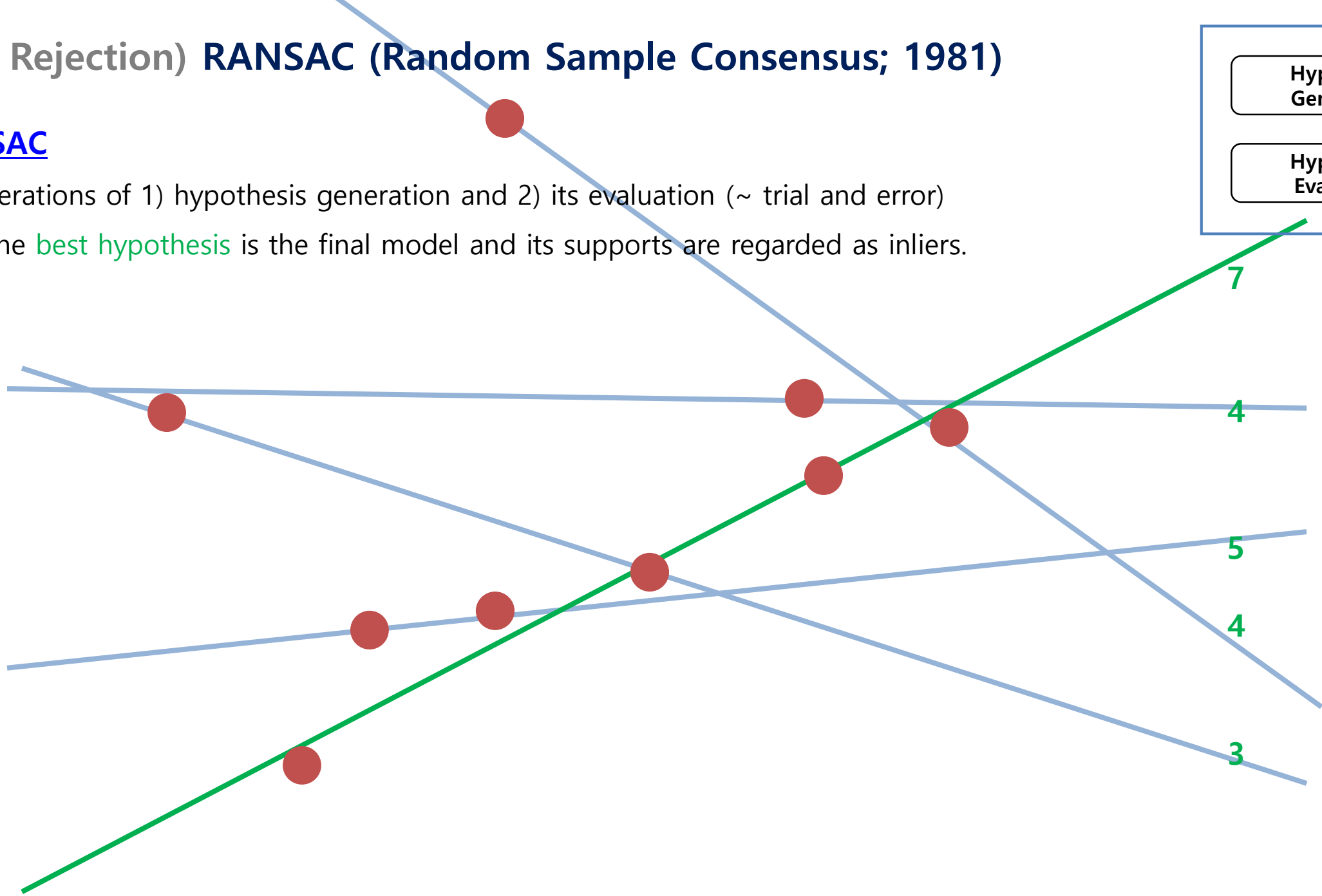
- **RANSAC**
  - Iterations of 1) hypothesis generation and 2) its evaluation (~ trial and error)
    - e.g. In this example, the 2D line is supported by points within the given threshold τ.



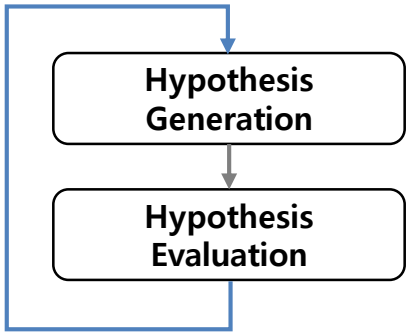
Outlier Rejection) RANSAC (Random Sample Consensus; 1981)

- **RANSAC**
  - Iterations of 1) hypothesis generation and 2) its evaluation (~ trial and error)
  - The best hypothesis is the final model and its supports are regarded as inliers.





Outlier Rejection) RANSAC (Random Sample Consensus; 1981)



- **RANSAC**
  - Iterations of 1) hypothesis generation and 2) its evaluation (~ trial and error)
  - **Q) How many iterations t are required?**
    - Parameters and assumptions
      - s: Success probability (confidence level)
      - d: The number of samples for model generation
      - γ: Inlier ratio
    - Success criteria: **Selecting d samples from inliers within t iterations**
      - Failure sampling: $1-\gamma^d$
      - Failure probability: $1-s=\left(1-\gamma^d\right)^t$
    - **The minimum number of iterations for success**: $t=\frac{\log(1-s)}{\log(1-\gamma^d)}$

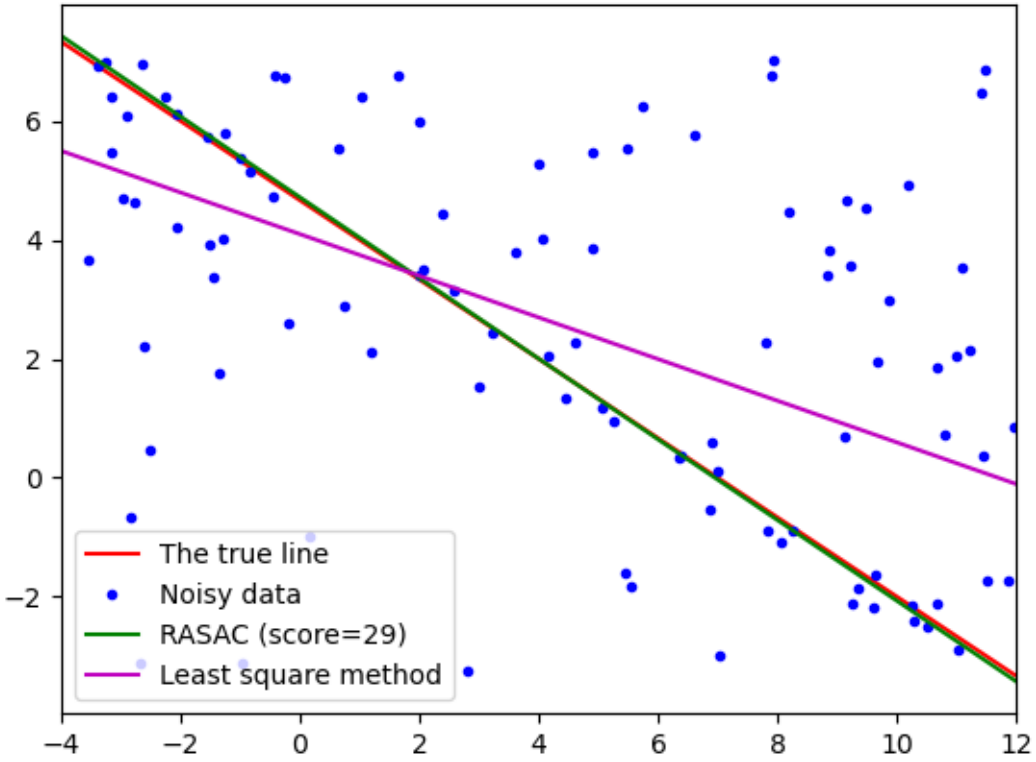
Outlier Rejection) RANSAC (Random Sample Consensus; 1981)

- Example) **Line fitting with RANSAC** [line_fitting_ransac.py]
  - Model definition: 2D line $\mathbf{m} = [a, b, c]^{\mathrm{T}}$ $(ax + by + c = 0;\ a^2 + b^2 = 1)$
  - Model generation: Two points can generate to a line.
  - Model evaluation: Geometric distance $e(\mathbf{x}_i; \mathbf{m}) = \frac{ax_i + by_i + c}{\sqrt{a^2 + b^2}}$

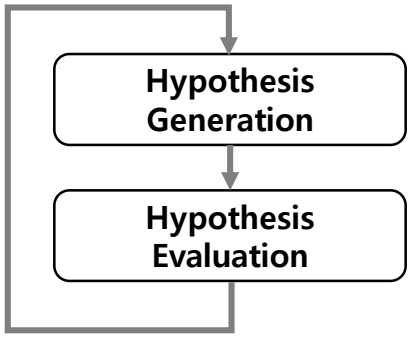
```
def generate_line(pts):
    # Line model: y = ax + b
    a = (pts[1][1] - pts[0][1]) / (pts[1][0] - pts[0][0])
    b = pts[0][1] - a * pts[0][0]

    # Line model: ax + by + c = 0 (a^2 + b^2 = 1)
    line = np.array([a, -1, b])
    return line / np.linalg.norm(line[:2])

def evaluate_line(line, p):
    a, b, c = line
    x, y = p
    return np.fabs(a*x + b*y + c)
```



Outlier Rejection) RANSAC (Random Sample Consensus; 1981)



- Example) **Line fitting with RANSAC** [line_fitting_ransac.py]

```
def generate_line(pts):
    ...

def evaluate_line(line, p):
    ...

def fit_line_ransac(data, n_sample, ransac_trial, ransac_threshold):
    best_score = -1
    best_model = None
    for _ in range(ransac_trial):
        # Step 1: Hypothesis generation
        sample = random.choices(data, k=n_sample)
        model = generate_line(sample)

        # Step 2: Hypothesis evaluation
        score = 0
        for p in data:
            error = evaluate_line(model, p)
            if error < ransac_threshold:
                score += 1
        if score > best_score:
            best_score = score
            best_model = model

    return best_model, best_score
```

Outlier Rejection) RANSAC (Random Sample Consensus; 1981)

- Example) **Line fitting with RANSAC** [line_fitting_ransac.py]

```
if __name__ == '__main__':
    true_line = np.array([2, 3, -14]) / np.sqrt(2*2 + 3*3) # The line model: a*x + b*y + c = 0 (a^2 + b^2 = 1)
    data_range = np.array([-4, 12])
    data_num = 100
    noise_std = 0.2
    outlier_ratio = 0.7

    # Generate noisy points with outliers
    line2y = lambda line, x: (line[0] * x + line[2]) / -line[1] # ax + by + c = 0 -> y = (ax + c) / -b
    y_range = sorted(line2y(true_line, data_range))
    data = []
    for _ in range(data_num):
        x = np.random.uniform(*data_range)
        if np.random.rand() < outlier_ratio:
            y = np.random.uniform(*y_range)        # Generate an outlier
        else:
            y = line2y(true_line, x)               # Generate an inlier
            x += np.random.normal(scale=noise_std) # Add Gaussian noise
            y += np.random.normal(scale=noise_std)
        data.append((x, y))
    data = np.array(data)

    # Estimate a line using RANSAC
    best_line, best_score = fit_line_ransac(data, 2, 100, 0.3) # log(1 - 0.999) / log(1 - 0.3^2) = 73

    # Estimate a line using OpenCV (for reference)
```

$$t > \frac{\log(1-s)}{\log(1-\gamma^d)} = \frac{\log(1-0.999)}{\log(1-0.3^2)} = 73$$

Review) Planar Homography

- Example) **Planar image stitching** [image_stitching.py]

```
# Load two images
img1 = cv.imread('../data/hill01.jpg')
img2 = cv.imread('../data/hill02.jpg')

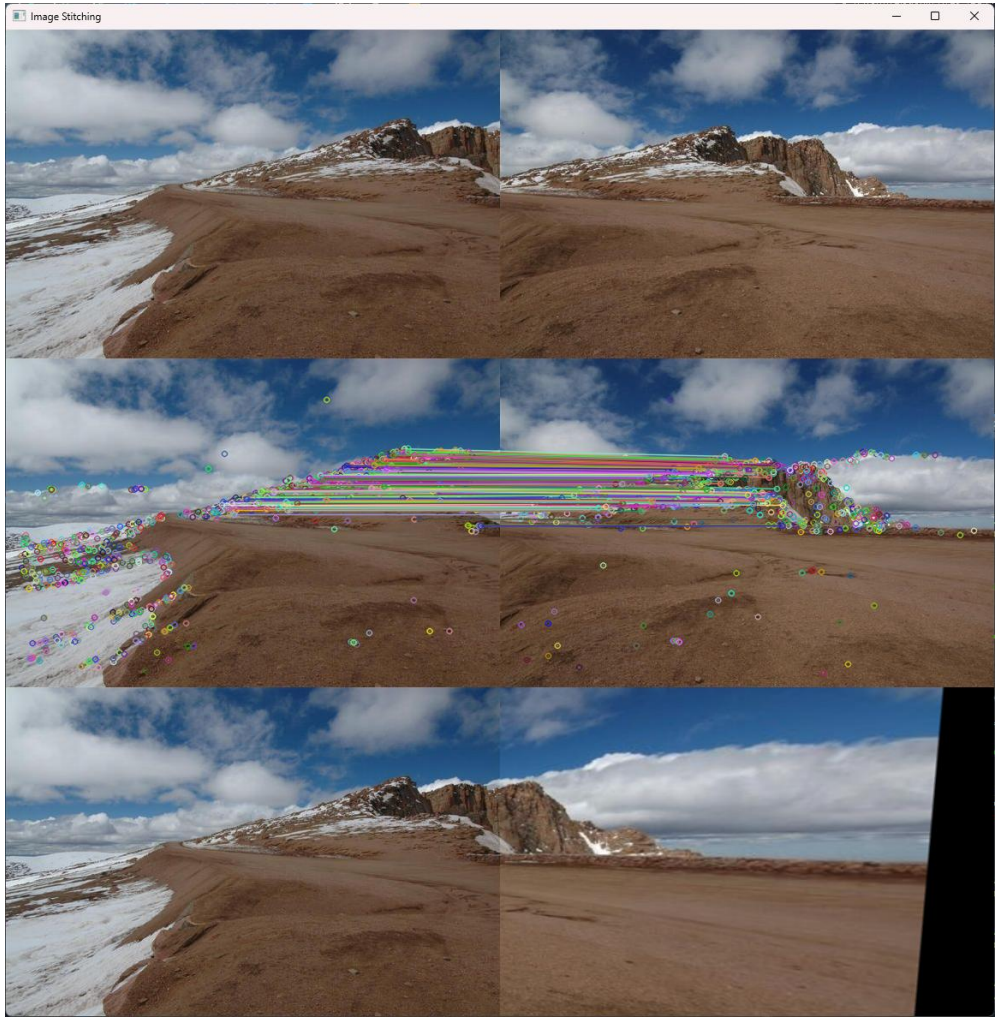
# Retrieve matching points
fdetector = cv.BRISK_create()
keypoints1, descriptors1 = fdetector.detectAndCompute(img1, None)
keypoints2, descriptors2 = fdetector.detectAndCompute(img2, None)

fmatcher = cv.DescriptorMatcher_create('BruteForce-Hamming')
match = fmatcher.match(descriptors1, descriptors2)

# Calculate planar homography and merge them
pts1, pts2 = [], []
for i in range(len(match)):
    pts1.append(keypoints1[match[i].queryIdx].pt)
    pts2.append(keypoints2[match[i].trainIdx].pt)
pts1 = np.array(pts1, dtype=np.float32)
pts2 = np.array(pts2, dtype=np.float32)

H, inlier_mask = cv.findHomography(pts2, pts1, cv.RANSAC)
img_merged = cv.warpPerspective(img2, H, (img1.shape[1]*2, img1.shape[0]))
img_merged[:,:img1.shape[1]] = img1 # Copy

# Show the merged image
img_matched = cv.drawMatches(img1, keypoints1, img2, keypoints2, match, None, None, None,
```



Outlier Rejection) RANSAC (Random Sample Consensus; 1981)

- Example) **Planar Homography estimation with RANSAC** [image_stitching_implement.py]

```
from homography_estimation_implement import getPerspectiveTransform
from image_warping_implement import warpPerspective2

def findHomography(src, dst, n_sample, ransac_trial, ransac_threshold):
    ...

if __name__ == '__main__':
    # Load two images
    ...

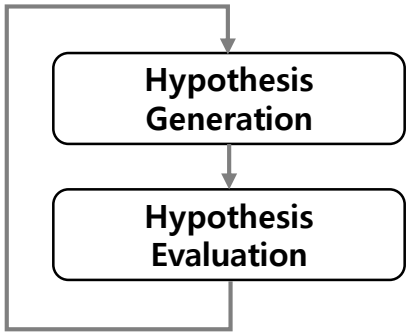
    # Retrieve matching points
    ...

    # Calculate planar homography and merge them
    ...

    H, inlier_mask = findHomography(pts2, pts1, 4, 1000, 2) # log(1 - 0.999) / log(1 - 0.3^4) = 849
    img_merged = warpPerspective2(img2, H, (img1.shape[1]*2, img1.shape[0]))
    img_merged[:,:img1.shape[1]] = img1 # Copy

    # Show the merged image
    img_matched = cv.drawMatches(img1, keypoints1, img2, keypoints2, match, None, None, None,
                                 matchesMask=inlier_mask) # Remove `matchesMask` if you want to show all ...
    merge = np.vstack((np.hstack((img1, img2)), img_matched, img_merged))
    cv.imshow(f'Planar Image Stitching with My RANSAC (score={sum(inlier_mask)})', merge)
    cv.waitKey(0)
```

Outlier Rejection) RANSAC (Random Sample Consensus; 1981)



- Example) **Homography estimation with RANSAC** [image_stitching_implement.py]

```python
def evaluate_homography(H, p, q):
    p2q = H @ np.array([[p[0]], [p[1]], [1]])
    p2q /= p2q[-1]
    return np.linalg.norm(p2q[:2].flatten() - q)

def findHomography(src, dst, n_sample, ransac_trial, ransac_threshold):
    best_score = -1
    best_model = None
    for _ in range(ransac_trial):
        # Step 1: Hypothesis generation
        sample_idx = random.choices(range(len(src)), k=n_sample)
        model = getPerspectiveTransform(src[sample_idx], dst[sample_idx])

        # Step 2: Hypothesis evaluation
        score = 0
        for (p, q) in zip(src, dst):
            error = evaluate_homography(model, p, q)
            if error < ransac_threshold:
                score += 1
        if score > best_score:
            best_score = score
            best_model = model

    # Generate the best inlier mask
    best_inlier_mask = np.zeros(len(src), dtype=np.uint8)
    for idx, (p, q) in enumerate(zip(src, dst)):
```

Note) $e(\mathbf{x}, \mathbf{x}') = \|\mathrm{H}\mathbf{x} - \mathbf{x}'\|_2$

Outlier Rejection) Least Squares Method vs. RANSAC

- **Least Squares Method**
  - Find a model while minimizing sum of squared errors, $\underset{\mathbf{m}}{\operatorname{argmin}} \sum_i e(\mathbf{x}_i; \mathbf{m})^2$
- **RANSAC**
  - Find a model while maximizing the number of supports (~ inlier candidates)

    ~ minimizing the number of outlier candidates

- **Q) Why is RANSAC robust to outliers?**
  - Problem formulation with a **loss function** ρ: $\underset{\mathbf{m}}{\operatorname{argmin}} \sum_i \rho\big(e(\mathbf{x}_i; \mathbf{m})\big)$
  - Loss function of least squares method: $\rho(x) = x^2$
  - Loss function of RANSAC: $\rho(x) = \begin{cases} 0 & \text{if } |x| < \tau \\ 1 & \text{otherwise} \end{cases}$

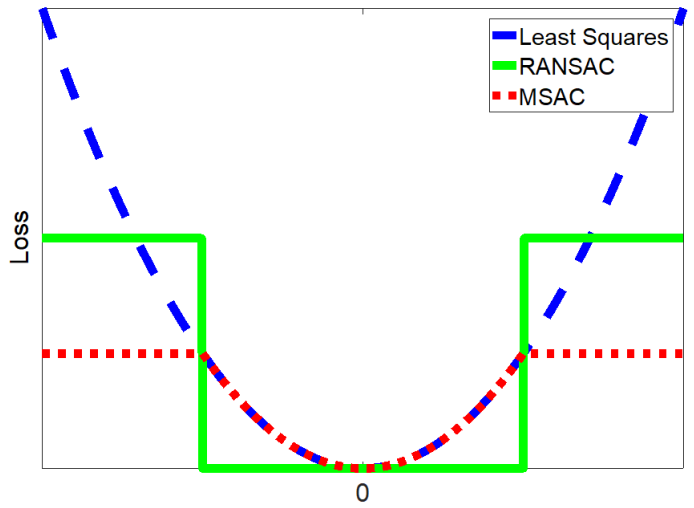
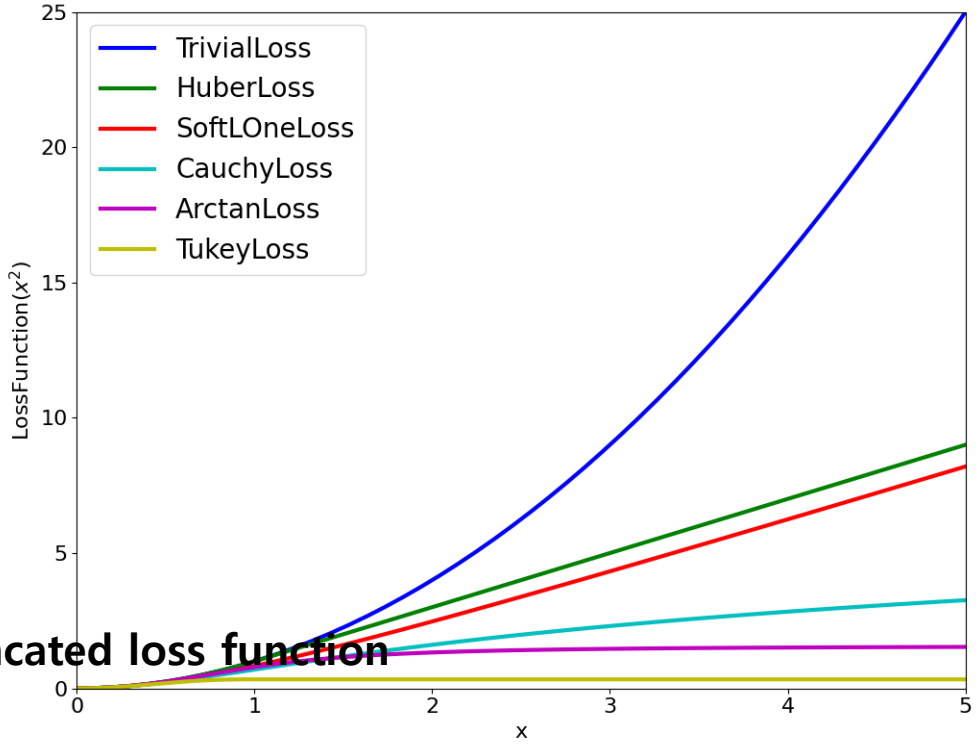


Image: Choi et al., "Performance Evaluation of RANSAC Family", BMVC, 2009 PDF

Outlier Rejection) M-estimator and MSAC

- **Robust regression**
  - Problem formulation with a **loss function** ρ: $\underset{\mathbf{m}}{\operatorname{argmin}} \sum_i \rho\left(e(\mathbf{x}_i; \mathbf{m})\right)$
  - Loss function of least squares method: $\rho(x) = x^2$
  - Loss function of RANSAC: $\rho(x) = \begin{cases} 0 & \text{if } |x| < \tau \\ 1 & \text{otherwise} \end{cases}$
- **M-estimator** (~ weighted least squares)
  - Find a model while minimizing sum of (squared) errors **with a truncated loss function**
  - Huber loss function: $\rho(x) = \begin{cases} x^2 & \text{if } |x| < \tau \\ \tau(2|x| - \tau) & \text{otherwise} \end{cases}$
  - Cauchy loss function: $\rho(x) = \ln(1 + \frac{x^2}{\tau^2})$
- **MSAC** (RANSAC implementation in OpenCV)
  - Consider inliers *with their quality* and regard the outliers *equally with limitation*
  - Loss function of MSAC: $\rho(x) = \begin{cases} x^2 & \text{if } |x| < \tau \\ \tau^2 & \text{otherwise} \end{cases}$



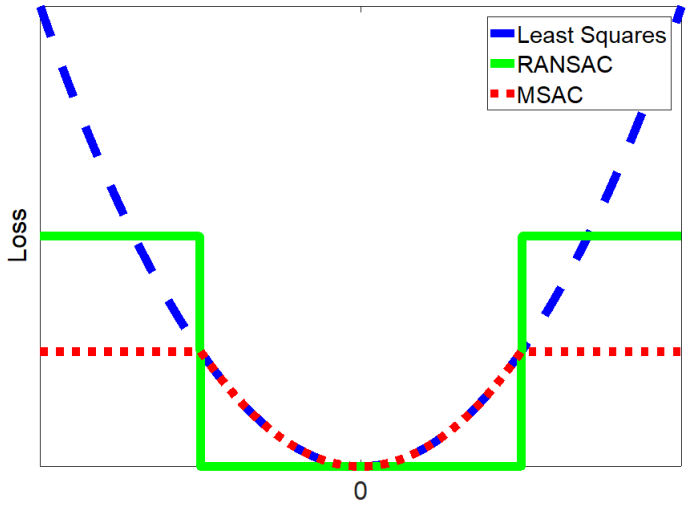


Image: Ceres Solver

Review) Solving Nonlinear Equation using Nonlinear Optimization

- **Example) Line fitting from more than two points such as $(1, 4)$, $(4, 2)$, and $(7, 1)$, using SciPy**
  - Unknown: Line parameters a, b, and c (line representation: $ax + by + c = 0$)
  - Cost function: $f(a, b, c) = \sum_i \left(\frac{ax_i + by_i + c}{\sqrt{a^2 + b^2}} \right)^2$
  - Optimizer: Gauss-Newton method (least squares)

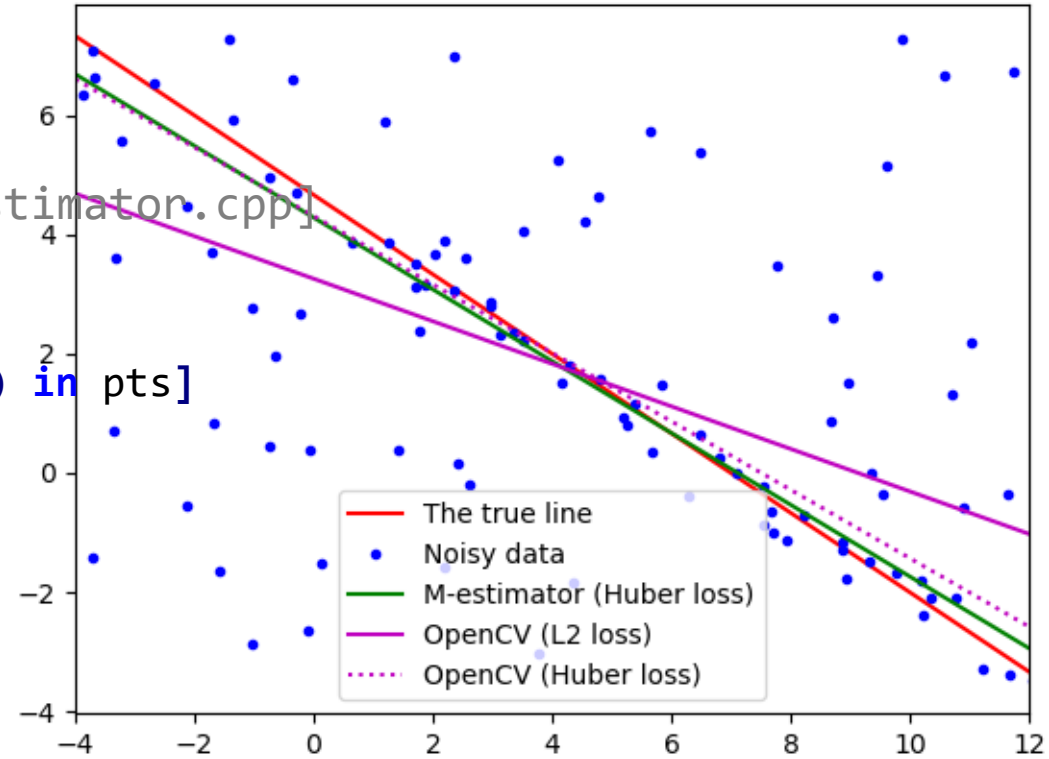
```
import numpy as np
from scipy.optimize import least_squares

def geometric_error(line, pts):
    a, b, c = line
    err = [(a*x + b*y + c) / np.sqrt(a*a + b*b) for (x, y) in pts]
    return err

pts = [(1, 4), (4, 2), (7, 1)]
line_init = [1, 1, 0]
result = least_squares(geometric_error, line_init, args=(pts,))
line = result['x'] / -result['x'][1] # [-0.50372575, -1., 4.34823633]
```

Outlier Rejection) M-estimator

- Example) **Line fitting with M-estimator** [line_fitting_m_estimator.cpp]




```
def geometric_error(line, pts):
    a, b, c = line
    err = [(a*x + b*y + c) / np.sqrt(a*a + b*b) for (x, y) in pts]
    return err

if __name__ == '__main__':
    # Generate noisy points with outliers
    ...

    # Estimate a line using least squares with a robust kernel
    init_line = [1, 1, 0]
    result = least_squares(geometric_error, init_line, args=(data,), loss='huber', f_scale=0.3)
    mest_line = result['x'] / np.linalg.norm(result['x'][:2])

    # Estimate a line using OpenCV (for reference)
    # Note) OpenCV line model: n_y * (x - x_0) = n_x * (y - y_0)
    nnxy = cv.fitLine(data, cv.DIST_L2, 0, 0.01, 0.01).flatten()
    lsqr_line = np.array([nnxy[1], -nnxy[0], -nnxy[1]*nnxy[2] + nnxy[0]*nnxy[3]])
    nnxy = cv.fitLine(data, cv.DIST_HUBER, 0, 0.01, 0.01).flatten()
    huber_line = np.array([nnxy[1], -nnxy[0], -nnxy[1]*nnxy[2] + nnxy[0]*nnxy[3]])
```

Outlier Rejection) Applications (Model Size)

- **Bottom-up approaches (~ voting)**
  - ~~**Hough transform**~~ (not suitable for more than three model parameters)
    - **A single datum** votes multiple model candidates.
      - Note) The parameter space is maintained as a multi-dimensional histogram (discretization).
    - Score: The number of hits by data
  - **RANSAC family**
    - **A sample of data** votes a single model candidate.
    - Score: The number of supports (whose error is within threshold)
  - Note) Application examples (small-size model): Line fitting, homography estimation, relative pose estimation, PnP
- **Top-down approaches**
  - **Optimization with truncated loss functions (e.g. M-estimator)**
    - The initial model parameter moves along with the gradient of the given cost function with **whole data**.
    - Score: A cost function with a truncated loss function.
  - Note) Application examples (large-size model): Camera calibration, visual SLAM, SfM

Image: pixabay

Summary) Finding Correspondences

- **Feature Points**
  - Gradient-based: Harris corner, GFTT corner, SIFT, SURF, …
  - Intensity-based: FAST, …
  - DL-based: LIFT, SuperPoint, …
- **Feature Descriptors**
  - Real-valued: SIFT, SURF, … (DL-based: LIFT, SuperPoint, …)
  - Binary-valued: BRIEF, ORB, …
- **Feature Matching**
  - Distance measures: Euclidean distance, cosine similarity, Hamming distance, …
  - Matching methods: Brute-force search, ANN search, …
  - DL-based: SuperGlue, …
- **Feature Tracking**
  - KLT tracker = GFTT detector + LK optical flow
- **Outlier Rejection (Robust Regression)**
  - RANSAC → MSAC (inlier quality)
  - Least squares method → M-estimator (robust kernels)

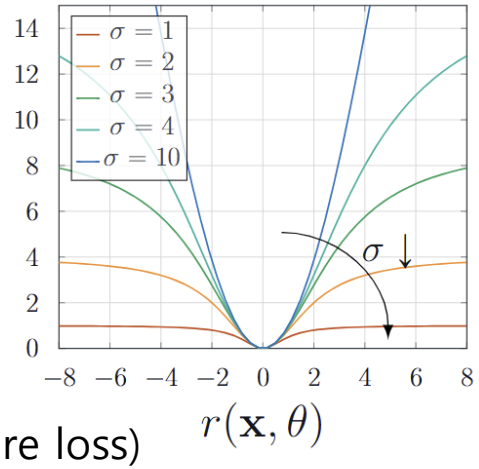


Beyond M-estimator with the fixed bandwidth threshold

- Adaptive or varying bandwidth thresholds (σ)
  - e.g. Graduated Non-Convexity (GNC)
    - Parameter estimation

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^{N} \rho_\sigma(r(\mathbf{x}_i, \theta))$$

      - θ: Parameters to estimate
      - ρ_σ : A robust loss function (e.g. Geman-McClure loss)
      - $r(\cdot)$: A residual function
    - Threshold update: $\sigma_k = \frac{\sigma_{k-1}}{\gamma_{\text{GNC}}}$
      - σ_k: The k-th bandwidth threshold
      - γ_{GNC}: A fixed annealing factor



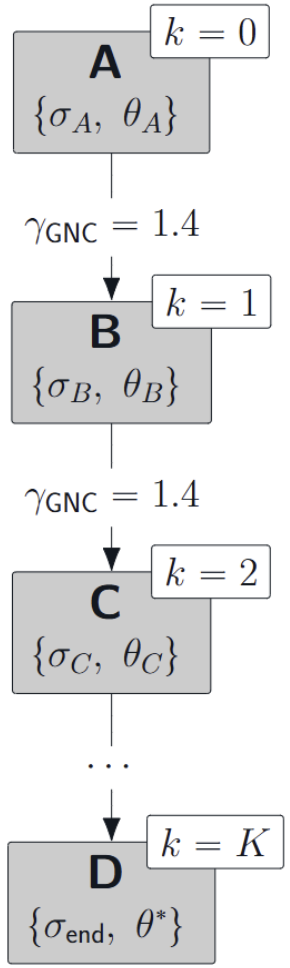


Image: Piedade et al., "SAC-GNC: SAmple Consensus for adaptive Graduated Non-Convexity", ICCV, 2025 Homepage CVF Github